

Do the gains from digital technology waves differ by income level? A systematic review, a pilot meta-regression and a macro-panel analysis

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Draft of 3 October 2026

Draft status (remove before submission). All study data in this draft were extracted by machine from the full texts. Every printed number used in the meta-regression was checked against the text twice, and 64 of the 67 study records were re-read once by a separate model run, and the author has since checked the extracted data by hand against the source papers [**author: state how many studies and entries were checked, how they were chosen, and what corrections were made**]. Screening used one assisted reviewer and no second screener. The protocol was not registered before the search was run. Scopus and Web of Science have not been searched. Sections marked [**author**] need the author's input. Numbers in this draft come from the files in the project repository, and the repository is the reference if text and data disagree.

Abstract

Background. Mobile telephony, the internet and, more recently, artificial intelligence (AI) are often expected to widen or narrow income gaps between countries, but the empirical cross-country evidence has not been synthesised across outcomes. **Methods.** We ran a systematic review following PRISMA 2020, searching OpenAlex, Crossref and NBER for 1995 to 2026, and assessed 107 full texts. We extracted effects on six outcome families, rated risk of bias with an adapted ROBINS-I tool, synthesised results narratively, and ran a pilot meta-regression of partial correlations for ICT-type adoption and output or productivity. For questions the studies cannot answer, we analysed World Bank data for 1995 to 2019 with fixed-effects panels, a staggered event study with not-yet-treated controls and local projections. **Results.** Sixty-seven studies met the criteria (54 on ICT, mobile, internet and broadband; 11 on AI; 2 on both). None was rated at low risk of bias (moderate 8, serious 50, critical 9). Most studies report a positive association between ICT-type adoption and output, but the pooled partial correlation across 16 studies is 0.19 (95% CI 0.10 to 0.27) with I^2 of 93%, and falls to between 0.005 and 0.06 after correcting for small-study bias. Estimates from GMM are on average 0.16 lower than those from other estimators, although restricting to studies with an identification strategy does not lower the pooled value. Twenty-six studies compare income groups or regions: eight find larger effects in richer economies, six in poorer ones, and three find no difference, mostly without a formal test. Returns are conditional on human capital, financial depth, governance and infrastructure. Evidence on AI comes from 13 studies, mostly short and using proxies. In the World Bank panel, mobile adoption was associated with faster growth in every income group under a same-period specification,

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most strongly in lower-middle income economies, but no income group showed a positive association under all four designs (same-period and predetermined panels, a staggered event study and local projections). **Conclusions.** The evidence does not establish that rich or poor economies gain more, and it cannot support causal statements. Credible identification for low- and middle-income economies is the main gap.

Keywords: digital technology; ICT; artificial intelligence; income level; systematic review; meta-regression; event study.

JEL codes: O33, O47, O11, L96.

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1 Introduction

Whether new technologies narrow or widen income gaps between countries is a long-standing question, and it has returned with artificial intelligence. Early empirical work found sizeable output effects of telecommunications infrastructure [1]. Later work on the long-run diffusion of technologies finds that technologies arrive at similar times in rich and poor countries, but that the intensity of their use diverges [2, 3]. If so, the gains from a new technology should depend on a country’s conditions at the time of adoption, and not only on whether the technology has arrived.

Evidence on this is scattered. Studies use different technologies (mobile subscriptions, internet users, composite ICT indices, ICT capital, robots, AI readiness indices), different outcomes (growth, productivity, employment, inequality, human development, emissions), different samples and different estimators. Individual studies often report one average effect and one or two interaction terms. A reader who wants to know whether a low-income economy can expect the same payoff as a high-income one has no single place to look, and a policymaker who reads one study can reach the opposite conclusion from one who reads another.

We therefore ask the following main question: *in empirical cross-country studies of digital and general-purpose technology waves, do the effects on output, structural change, labour markets, distribution, living standards and resource use differ by income level and by country characteristics, how large are the differences, and how credible is the evidence?* We keep studies of historical waves (ICT, mobile, internet, broadband, crypto-assets) separate from studies of AI, because AI evidence is younger and thinner. We also ask a set of subsidiary questions that the data we collected allow, and four questions that the published studies cannot answer, which we address with World Bank data.

The paper has three parts. The first is a systematic review that follows PRISMA 2020, with a structured extraction of effects, a quality rating of each study and a narrative synthesis across six outcome families. The second is a pilot meta-regression of effect sizes for the best-covered question, the association between ICT-type adoption and output or productivity, together with nine exploratory questions about measurement, study design and the conditions under which returns differ. The third is an original analysis of World Bank panel data that gives a first, transparent answer to four questions about the shape of the effect by adoption level, effects by income group under one design, timing around technology arrival, and differences inside low-income economies.

The contribution is a map of the evidence with an explicit quality rating of each study, a transparent count of where income-group comparisons exist and where they do not, an estimate of how much of the apparent positive association survives bias corrections, and a panel analysis whose limits are stated at each step. We are explicit about what the data cannot show. All data and code are in the project repository.

The rest of the paper is organised as follows. Section 2 sets out the conceptual background and the hypotheses. Section 3 describes the methods. Section 4 reports the review and the meta-regression, and Section 5 reports the panel analysis. Section 6 discusses the findings, and Sections 7 and 8 give the limitations and conclusions.

2 Conceptual background and hypotheses

2.1 Why the same technology could pay differently

Four mechanisms recur in the literature on information and communication technology and on automation, and each implies a reason why the gain from a technology might differ by income level.

Infrastructure and complementary investment. Telecommunications infrastructure raises output, and the effect may depend on how much infrastructure already exists, because network benefits grow with the size of the network [1]. Where electricity, roads and financial services are scarce, the same handset or connection may yield less. This implies that the gain depends on a country's stock of complementary assets, which is usually lower in poorer economies.

Information and market access. Mobile phones can reduce price dispersion and waste where markets are thin. In a study of South Indian fisheries, the arrival of mobile phones reduced price dispersion and waste and raised welfare [4], and a review of evidence from Africa describes how mobile phones can improve information flows in agricultural and other markets [5]. These mechanisms are strongest where information was previously scarce, which is more common in lower-income settings, and they imply large early gains at low adoption levels.

Skills and tasks. Computers substitute for routine tasks and complement non-routine tasks [7], and automation both displaces and creates tasks [8]. In Africa, the arrival of fast internet through submarine cables was associated with higher employment, and the effect was larger for more skilled workers [6]. The gains therefore depend on the skills of the workforce, which differ by income level.

Absorptive capacity. The ability to use external knowledge depends on prior related knowledge [9]. For technology adoption this means that returns depend on human capital, management capacity and institutions, again variables that differ systematically by income.

Two forces therefore pull in different directions. Catch-up and early information gains suggest that low-adoption, lower-income economies could gain more per unit of adoption. Complementary assets, skills and absorptive capacity suggest that higher-income economies could gain more. A third consideration is time: the long-run diffusion evidence suggests that gains may build over years, and a short window may miss them [3]. A review of the empirical evidence on information technology and economic performance, published early in the period we study, describes the findings as mixed [10]; it is not an estimating study and is not among our included studies.

2.2 Hypotheses and questions

These mechanisms lead to the following hypotheses, which structure the review and the panel analysis. They are stated so that the data can contradict them.

H1 ICT-type adoption is positively associated with output and productivity, but the size of the association is modest and varies strongly across studies.

H2 The association is larger where complements are stronger: higher human capital, deeper financial markets, better governance and better infrastructure.

- H3** The association differs by income level, but its sign is not predicted, because catch-up and complement effects have opposite implications.
- H4** Estimates from designs that address reverse causation are smaller than those from designs that do not.
- H5** Gains depend on timing, appearing with a lag, so that studies measuring only contemporaneous effects understate them.
- H6** Evidence on AI follows the same pattern as evidence on earlier technologies but is thinner and of lower quality.

3 Methods

3.1 Protocol and registration

The review follows PRISMA 2020 [11] and the structure of PRISMA-P [12]. The protocol (version 1.0, 2 October 2026) is in the repository. It was intended to be registered on the Open Science Framework before the search, but registration had not been completed when the search was run. This is a deviation from good practice, and we log the amendments made after the search in Appendix H. **[author: register the protocol on OSF and state the registration number or explain]**

3.2 Eligibility criteria

We included quantitative empirical studies with their own estimation (panel regression, instrumental variables, growth accounting, event studies, structural estimation) that relate adoption, diffusion, investment or capability of ICT in general, mobile telephony, internet or broadband, crypto-assets or AI to at least one outcome in six families: (1) output and productivity, (2) structural change, (3) labour market, (4) poverty and distribution, (5) living standards beyond income (consumption, health, schooling, financial inclusion) and (6) resource cost (electricity use, carbon intensity). Historical-wave studies needed at least 10 economies or an explicit income-group comparison; AI studies needed at least 3 economies or an income-level comparison. Studies from 1995 onward in English were eligible. Purely conceptual papers, reviews and studies whose only outcome was adoption itself were not synthesised. Full criteria and exclusion codes are in Appendix D.

3.3 Information sources and search

We searched OpenAlex (a Boolean query restricted to the economics field), Crossref (96 technology-by-context queries) and the NBER working-paper series, covering 1995 to the search date of 2 October 2026. The search blocks and the OpenAlex query are in Appendix C. arXiv was attempted but did not respond in usable time. Scopus, Web of Science and EconLit were not searched, because they need institutional access. After removing duplicates by DOI and by normalised title and year, 3,612 records remained.

3.4 Screening and full-text access

Screening was done by one assisted reviewer. Automated rules removed records with no eligible technology term, no outcome term or no multi-economy cue (stage 1). Title and abstract screening then applied the eligibility table, with four holding categories for records that were not eligible but useful (context on diffusion, environmental-only outcomes, reviews for citation search and supplementary micro evidence). No second screener was available, so we report no agreement statistic. A random verification sample of 90 decisions is in the repository awaiting an independent check.

Full texts were obtained from open-access sources, repositories and the author. Records without a free full text, and those the author judged to come from unreliable venues after visiting the journal site, were not read and are not used as evidence. This creates a selection mechanism that favours open-access studies; Section 7 discusses it. Three studies were added late (71, 278 and 341) after the author supplied them; one further study (89) was supplied and excluded.

3.5 Data extraction

For each included study we extracted technology, exposure measure, sample, estimator, identification strategy, results by outcome family with printed estimates and uncertainty, heterogeneity findings, and group-specific results, using a fixed form (Appendix D). Extraction was done by a language-model agent working from converted full texts, with a locator and a short verbatim quote for every result. An automatic check matched 403 of 586 quotes verbatim across all 107 records read (the remainder are affected by table layout and paraphrase).

3.6 Second-pass check

A second, separate model run then re-read each of 64 included records and each of 203 regression estimates against the text and classified every item as confirmed, corrected or unverifiable. This catches transcription and reading errors. It is not an independent check: the same family of model reads the same machine-converted text, and garbled tables stay garbled. The three studies added late (71, 278 and 341) were not given this second pass; we checked their key printed numbers by text search. Appendix F summarises the second pass by study. Finally, the author checked the extracted data by hand against the source papers. **[author: scope of the hand check (number of studies and entries, selection method), number of discrepancies and how they were resolved]**

3.7 Risk of bias

Risk of bias used a six-domain tool adapted from ROBINS-I [14]: confounding and reverse causation, exposure measurement, selection of countries and years, outcome measurement, specification and researcher degrees of freedom, and selective reporting. The overall rating was low, moderate, serious or critical, with the definitions in Appendix D. One rater made the first-pass ratings; the second pass offered an opinion on each, which we report without changing the ratings.

3.8 Narrative synthesis

We synthesised results narratively, following the SWiM reporting guideline [13], by outcome family and by technology stream. Studies were grouped by outcome family; within each family we tabulated the direction of the results, the estimator, the sample and the risk of bias (Appendix B). Counts of study directions are descriptive; they are not weighted by sample size or quality and are not effect sizes.

3.9 Pilot meta-regression

For ICT-type adoption and output or productivity we ran a pilot meta-regression in the tradition of Stanley and Doucouliagos [15]. Each estimate was converted to a partial correlation coefficient, $PCC = t/\sqrt{t^2 + df}$, with standard error $\sqrt{(1 - PCC^2)/df}$ and df equal to the number of observations minus 10 regressors (a sensitivity check uses 0, 5 and 20). The t -statistic came from the printed coefficient and standard error, or from a printed t -statistic. Estimates reported with p -values or intervals only were not converted. We excluded interaction terms, crypto and AI exposures, estimates without a usable number of observations, and five estimates judged not poolable on the second pass (a fintech-credit exposure, a growth outcome mixed with level outcomes, and a TFP index in levels). We pooled one headline estimate per paper by random effects [17], ran a weighted moderator regression with standard errors clustered by paper, and tested for publication selection with the funnel-asymmetry and precision-effect tests (FAT-PET) and the precision-effect estimate with standard error (PEESE) [16]. With 18 papers, the clustered standard errors are optimistic and p -values are indicative.

3.10 Additional exploratory questions

Beyond the main question we specified nine further questions and their analyses before running them (`review/extra_questions_spec.md`), all using the data already collected: whether the pooled association depends on the measurement of the technology (a composite index versus a single indicator), on the outcome form (growth versus level) and on the sample period (added one at a time to the meta-regression); whether weaker designs are more common where low-income economies are sampled; whether study quality goes with the reported direction; which enabling conditions authors test and in which direction (136 heterogeneity entries classified by one model reader, with a second reader on a random 20% sample); the thresholds stated; an evidence-gap map by outcome family and income group; and a comparison of the AI and historical-wave streams. These analyses are exploratory, are not corrected for multiple testing, and should be read as hypotheses.

3.11 Macro-panel analysis

Four questions (the shape of the effect by adoption level, effects by income group under one design, timing around technology arrival, and differences inside low-income economies) cannot be answered from the included studies. We therefore specified, before downloading the data, an analysis of public World Bank World Development Indicators for 1995 to 2019 (`analysis/spec.md`), and later amended the event-study design (`analysis/spec_amendment.md`) after the first design failed its own checks. The variables and their World Bank codes are in Appendix G.

Fixed-effects panels (A1 to A3, A6). In five-year periods (1995–2000, 2000–05, 2005–10, 2010–15 and 2015–19), annual growth of GDP per capita was regressed on the change in mobile subscriptions per 100 people over the same period (A1) or the previous period (A2), with initial log income, initial gross secondary enrolment, the change in trade openness and period effects, separately by current World Bank income group, with standard errors clustered by country. A3 splits the change by baseline adoption level, and A6 adds enabling conditions inside low and lower-middle income economies, one moderator at a time.

Event study (A4b). The takeoff year is the first year mobile subscriptions reach 10 per 100 people among economies below that level in 1995. For each cohort of economies with the same takeoff year and each horizon from five years before to ten years after, the effect compares the change in log GDP per capita from the year before takeoff with the change in economies not yet treated at that horizon, after adjusting for initial income and earlier growth by outcome regression, in the manner of Callaway and Sant’Anna [18]. Estimates are averaged across cohorts, weighted by the number of treated economies, and intervals come from a bootstrap over economies. Effects at horizons before takeoff are placebo tests, and a subgroup is interpreted only if the interval of their mean includes zero. We use not-yet-treated controls because standard two-way fixed-effects event studies can be biased when treatment is staggered and effects vary over time [19, 20].

Local projections (A7). For a design without a binary treatment, we estimated local projections [21] on the annual panel: cumulative growth of GDP per capita from year t to $t + h$ on the change in mobile subscriptions over the previous three years, with country and year fixed effects, controls for earlier growth and initial income, and separate coefficients by income group.

All of these are associations. We have no instrument, so we do not claim a causal effect, and cost-effectiveness cannot be examined with these data.

4 Results of the review and the meta-regression

4.1 Study selection

Figure 1 and Table 1 give the flow. Of 247 records sent to full-text assessment, 106 full texts were obtained and read, together with one review. After the second pass, 67 studies are included, 39 were excluded at full text (12 for too few economies, 5 for an ineligible exposure, 1 for an ineligible outcome, 21 for being conceptual or lacking their own estimation) and one was kept as context. Three studies (298, 878 and 1495) were moved from include to exclude after the second pass on eligibility grounds; this should be confirmed by a human reviewer. About 140 of the 247 records have not been read because no free full text was available.

4.2 Characteristics of included studies

The 67 included studies were published between 2003 and 2026, and 52 of them from 2020 onward (Figure 2). Fifty-four study ICT, mobile, internet or broadband; 11 study AI; 2 cover both. Most use country-year panels (48), with a few at industry-country level. In a rough classification of the main estimator, 25 use system or difference GMM, 24 use fixed-effects, random-effects or pooled OLS, 11 use ARDL, cointegration or heterogeneous-panel estimators

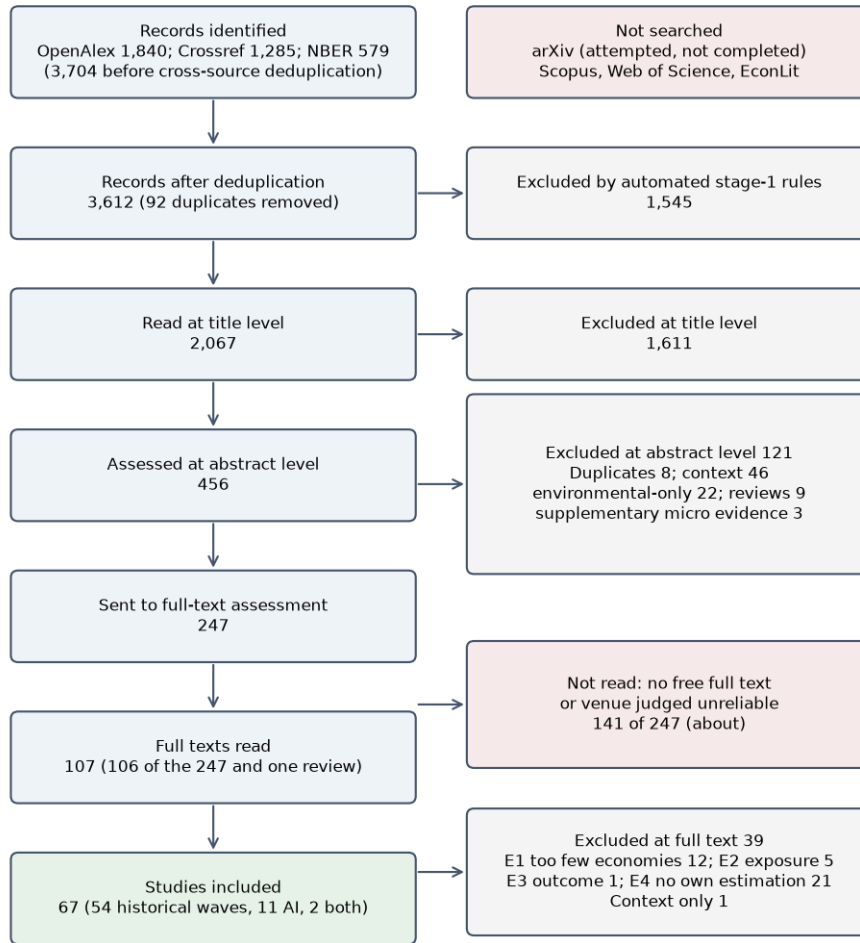


Figure 1: PRISMA 2020 flow diagram.

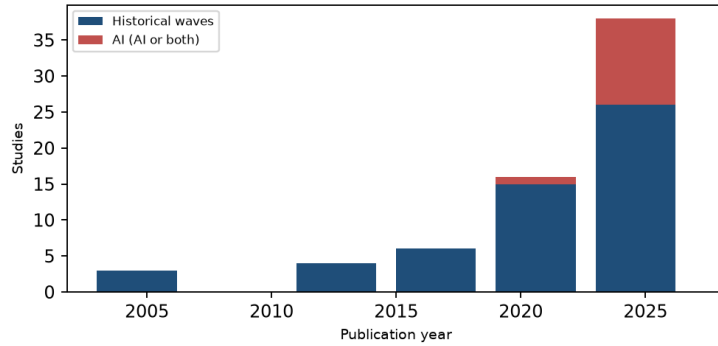
and 7 use growth accounting, data envelopment or other methods. Thirty-six studies have no identification strategy beyond controls and fixed effects. Twenty list at least one low-income economy; studies that do not state their income coverage are counted as not listing one. Outcome families, with a study able to contribute to several, are: output and productivity 37, labour market 14, living standards beyond income 12, poverty and distribution 11, structural change 5, resource cost 4. Appendix A lists all studies and Appendix B gives the results by family.

4.3 Risk of bias

No study was rated at low risk of bias. The first-pass ratings were moderate for 8 studies, serious for 50 and critical for 9 (Figure 3). In the second pass, the separate reader agreed with the overall rating for 55 of 64 studies, found six too harsh (268, 302, 323, 408, 707 and 1469) and three too lenient (57, 957 and 1031). We did not change any rating, and these differences should be settled by a human reviewer. The commonest weaknesses are adoption treated as exogenous, indices whose scope does not match the technology, and many specifications with no stated preferred one.

Table 1: Study selection (counts from the project repository).

Stage	n
Records identified (OpenAlex 1,840; Crossref 1,285; NBER 579; before cross-source deduplication 3,704)	3,704
Duplicates removed across sources	92
Records after deduplication	3,612
Excluded by automated stage-1 rules	1,545
Read at title level / excluded at title level	2,067 / 1,611
Assessed at abstract level / excluded at abstract level	456 / 121
Held as duplicates, context, environmental-only, reviews, supplementary	88
Sent to full-text assessment	247
Full texts read (106 of the 247 plus one review)	107
Excluded at full text	39
Kept as context only	1
Included	67

**Figure 2:** Included studies by publication year and stream.

4.4 Accuracy of the extraction

Across 372 result entries in the 64 re-read records, 348 were confirmed, 19 corrected and 5 could not be verified; 13 of 133 heterogeneity entries were corrected. Of 203 regression estimates, 202 matched the printed tables and one needed a corrected income label. Corrections were mostly misread labels, counts, and claims that the source text does not make. They were applied or flagged in the repository. Because both passes used the same family of model reading the same machine-converted text, the author’s hand check against the source papers is therefore the reference. [author: report agreement and corrections]

4.5 Output and productivity

Thirty-seven studies cover output or productivity (4 on AI). Most report a positive association, but the sign is not uniform: current and lagged mobile effects have opposite signs in one 41-economy GMM study (study 10); ten high-income countries show no significant effect of any ICT indicator (study 1448); and in 27 EU states the digitalisation coefficient loses significance once economic freedom is added (study 988). In 11 MENA economies, CS-ARDL estimates of an ICT index on GDP per capita are 0.05 (long run) and 0.06 (short run), both significant only at 10%, while the dynamic common correlated effects estimate is 0.30; the paper’s own causality tests show feedback in both directions (study 71). A digital economy index lowers green total

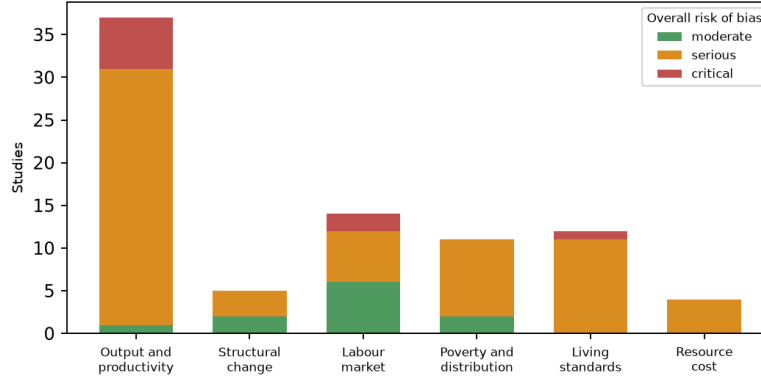


Figure 3: Overall risk of bias by outcome family (a study counts in each family it reports).

factor productivity across 40 Belt and Road economies (study 245), although that outcome sits closer to resource cost.

Interaction terms are the most repeated finding: ICT pays more where human capital is higher in two Asian GMM studies (302 and 323, which share authors, panel and index), a developing-country productivity study, and in Africa, where the effect of the digital economy is negative below a human-capital index of about 2.35 and the average country is below it. Financial development, trade openness, governance, electricity and the diffusion level of ICT also appear as moderators. These are conditional associations from single-equation panels and do not show that raising a moderator would change the return. A non-parametric study of 24 OECD economies finds that ICT capital change has no contemporaneous effect on the production frontier and a positive effect five years later, which supports the view that timing matters (H5). In the four AI studies, developed economies show about twice the productivity effect of emerging economies (0.24 against 0.11) in a 10-economy comparison rated at critical risk of bias; in 63 low and lower-middle income economies, AI-related imports are associated with higher labour productivity, more so where connectivity, electricity and financial inclusion are better; and two further studies (a sub-Saharan firm-level study and a three-country generative-AI survey) are rated critical and cannot be relied on.

4.6 Structural change

Five studies cover structural change. Internet penetration is associated with faster structural change in a 51-economy panel, more so in low and lower-middle income and agrarian economies than in upper-middle and high-income ones, and more so where forward value-chain linkages are higher; manufacturing employment falls in South Asia and is unchanged in sub-Saharan Africa. A 3G coverage study in 14 developing countries finds jobs created in services and non-farm self-employment but no shift of labour away from agriculture. ICT is positive for industry and services value added but not for agriculture in a 26-economy PMG study, although columns 2 to 4 of its Tables 10 and 11 are identical across industry and services, so that paper needs a human check. A study of about 100 developing economies finds that ICT lowers export concentration, with a larger effect in the low and lower-middle income group (-0.155 on the concentration index) than in the high and upper-middle group (-0.020), but no formal test of the difference. The evidence base here is still small.

4.7 Labour market

Fourteen studies cover labour markets, five of them on AI. In advanced economies ICT is linked to routine-task decline and polarisation, and capital-skill complementarity raises skill premia. In 27 European countries ICT capital creates more jobs where the share of high-skilled jobs is lower. No included study measures routinisation in low-income economies. Third-generation mobile coverage raises women's labour-force participation but not men's in 14 low- and middle-income countries, in one of the few studies that uses an instrument for developing-country labour outcomes. AI exposure is positively associated with employment growth, concentrated in high-education occupations and younger workers, in European samples; in Southeast Asia the pattern differs by country, though the country-level estimates could not be checked in the source figure. Across 29 countries, the AI index effect on employment declines with labour productivity level. A 78-economy readiness-index study is rated critical. The pattern across studies is that AI complements skills and displaces the rest, and the evidence on low-income economies is nil.

4.8 Poverty and distribution

Eleven studies cover poverty and distribution, two on AI. Results split. A threshold model for 45 developing countries finds that ICT raises the Gini coefficient below a digital-maturity score of 0.48 and lowers it above, an inverted U. In 48 sub-Saharan African economies ICT is positively associated with a composite inclusive-growth index (0.03 to 0.15 across specifications), with trade openness negative on its own and offset by ICT in the interaction, though the paper is inconsistent about its estimator. A study of poverty and inequality finds that the effect of inequality on poverty reverses above a technology threshold, mobile banking is mostly null or weakly negative on average inclusive growth with effects at the tails of the distribution, and robots raise carbon inequality between countries, less so in open and well-connected economies. The consistent message is that effects depend on how far a country has gone: below some level digitalisation widens the gap, above it narrows it. The thresholds are estimated in single studies with large uncertainty and have not been replicated.

4.9 Living standards beyond income and resource cost

Twelve studies cover living standards. Mobile subscriptions are positively associated with the Human Development Index in high-income and lower-middle-income groups and not in low-income countries in a large panel. ICT slopes on the index are positive in South Asia, sub-Saharan Africa and Latin America. In low and lower-middle income countries ICT narrows gender education gaps up to about 100 mobile subscriptions per 100 people, after which the gap widens again. Internet raises financial inclusion more where governance is better in sub-Saharan Africa, and mobile helps only where internet access is adequate. An AI development index raises human-capital-based educational development in developed economies and not in developing ones, with a sharp pandemic-period jump.

Four studies cover resource cost, two on AI. Digitisation is associated with higher carbon productivity, and the effect falls with income (0.085 for high-income, 0.058 for middle-income, 0.041 for low-income countries), while a digital-economy index lowers green total factor productivity in Belt and Road economies. Four studies are too few to generalise.

4.10 Do gains differ by income level?

Twenty-six studies compare income groups or regions, formally or descriptively (Table 2). Eight find larger effects in richer or OECD economies, six in poorer or less productive ones, and three find no clear difference; one finds a negative effect in all groups. Several of the first group have no test of the difference, and the studies finding larger effects in poorer economies include one single cross-section rated critical. The remaining studies compare groups without a directional finding, and study 268 appears in both of the first two groups because its direct and moderating effects point in different directions. We therefore treat the evidence as inconclusive. Studies of richer economies mostly use OLS or fixed effects, test the difference informally, and rarely include low-income economies, so the lean toward larger measured gains in richer samples may reflect study design.

Table 2: Studies that compare income groups or regions (study IDs in Appendix A).

Finding	Studies	Notes
Larger in richer or OECD	88, 138, 162, 268 (direct effect), 276, 535, 750, 957	Several lack a test of the difference; 535 rated critical and its figures look illustrative; 957 compares OECD with MENA
Larger in poorer or less productive	268 (informality moderation), 341, 407, 707, 952, 1192	407 compares Central and Eastern with Western Europe; 1192 is a single cross-section rated critical; 952 flips sign between pooled and subgroup models
No clear difference	29, 320, 879	879 compares low with lower-middle income only
Negative in all groups	245	Green total factor productivity outcome

4.11 Evidence on AI

The 13 studies on AI (11 AI only, 2 both streams) are mostly short panels of 3 to 78 economies, typically starting in 2012 or later, with exposure measured by occupational exposure indices, national readiness indices, patent-based measures or industrial robot density. None measures realised use of generative AI at country level. Where AI studies show an effect, it tends to be larger in richer economies and in high-skill occupations. The evidence from low-income economies is one productivity study of 63 low and lower-middle income economies and two studies rated critical. AI cannot be compared with earlier waves on equal terms.

4.12 Pilot meta-regression: ICT-type adoption and output

Of 206 estimates from 26 papers, 103 from 18 papers were usable. For the 16 papers with a convertible headline estimate (Figure 4), the random-effects pooled partial correlation is 0.189 (95% CI 0.104 to 0.274), with $\tau^2 = 0.027$, $I^2 = 93\%$ and a 95% prediction interval from -0.141 to 0.519 . A partial correlation of 0.19 corresponds to about 3.6% of residual variance. Of the headline estimates, 88% are positive and 75% are positive and statistically significant.

Table 3 shows that GMM estimates are on average 0.16 lower in partial correlation than estimates from other methods. This is consistent with reverse causation inflating simpler estimates, though it cannot prove it. A study-level check points the other way, however (see the sensitivity

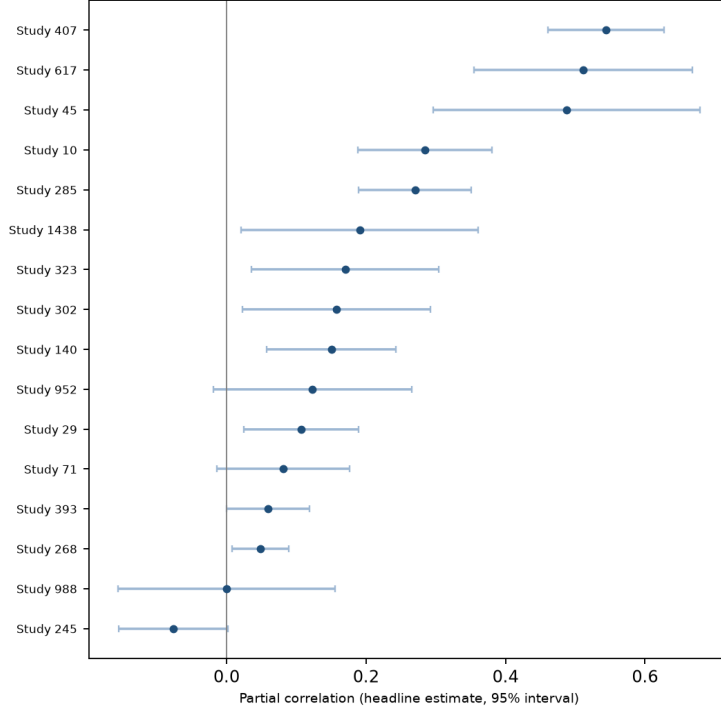


Figure 4: Headline partial correlations by study (95% intervals).

Table 3: Moderator meta-regression of partial correlations (103 estimates, 18 papers, standard errors clustered by paper).

Moderator	Coefficient	Clustered SE	p
Constant	0.087	0.056	0.12
Low- or middle-income sample	-0.038	0.045	0.40
System or difference GMM	-0.163	0.048	< 0.01
Mobile (relative to general ICT)	0.053	0.052	0.31
Internet or broadband (relative to general ICT)	0.135	0.079	0.09

analyses below), so H4 receives only mixed support. The low- or middle-income indicator is not significant, but only 5 papers (11 estimates) come from such samples, against 16 papers and 92 estimates for high-income or mixed samples (mean partial correlation 0.031 against 0.123, unweighted). The FAT-PET slope on the standard error is 1.91 ($p = 0.06$) with a bias-corrected intercept of 0.005 ($p = 0.83$), and the PEESE intercept is 0.064 ($p = 0.04$), so the pooled value may be substantially inflated by selective reporting (Figure 5). The pooled estimate is stable to the degrees-of-freedom assumption (0.186 to 0.193 for 0, 5 and 20 regressors).

Sensitivity analyses. Dropping each study in turn moves the pooled partial correlation between 0.159 and 0.208, so no single study drives the result. Excluding the studies rated critical gives 0.169 (15 papers, 95% CI 0.086 to 0.253), and excluding one of the two studies that share authors, panel and index (323) gives 0.191. Only one paper with a usable headline estimate was rated at moderate risk of bias, so a restriction to moderate-risk studies is not possible. Restricting to the 8 studies whose design includes an identification strategy beyond controls (internal or external instruments, lagged regressors or a timing design) gives a *higher* pooled value of 0.222 (95% CI 0.157 to 0.287, $I^2 = 57\%$), not a lower one. The estimate-level GMM coefficient and this study-level restriction therefore disagree, and we do not read either as

showing that better identification lowers the association. GMM estimates in these studies are often short-run effects of levels on growth rates, and units differ, which may account for part of the gap.

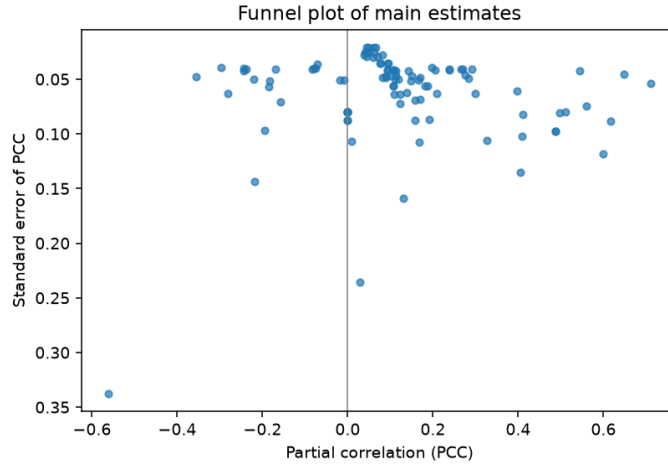


Figure 5: Funnel plot of main estimates (partial correlation against its standard error).

4.13 Exploratory analyses of additional questions

Measurement, outcome form and period. Adding single variables to the meta-regression, composite-index exposures show a partial correlation 0.07 higher than single indicators (SE 0.11, $p = 0.50$) and growth-rate outcomes 0.04 higher than level outcomes (SE 0.10, $p = 0.67$); neither is distinguishable from zero, and both rest on between-paper contrasts because few papers contribute both kinds. Later samples show lower partial correlations (-0.014 per year of sample midyear, SE 0.009, $p = 0.11$), which would be about -0.14 per decade. This is not significant, is partly confounded with the shift to GMM in recent studies, and is a hypothesis about diminishing returns, not a finding.

Design and income coverage. Studies that list a low-income economy were not more likely to lack an identification strategy (8 of 20, 40%, against 28 of 47, 60%; Fisher $p = 0.18$) or to be rated at higher risk of bias (moderate risk: 2 of 20 against 6 of 47, $p = 1.00$). We therefore find no evidence that weaker designs cluster where low-income economies are sampled, although the comparison has little power. The test of whether study quality goes with a positive direction for output could not be run: only one of the 37 output studies was rated at moderate risk of bias.

Enabling conditions. Of the 136 heterogeneity entries, 105 are moderation tests. Human capital is tested in 14 studies (amplifies the effect in 8, dampens it in 3, mixed in 4), income group in 18 (richer or higher-income group larger in 7, smaller in 3, mixed in 3, no difference in 4, unclear in 2), infrastructure or connectivity in 8 (5 amplify), institutions or governance in 8 (3 amplify, 2 dampen, 4 mixed), trade openness in 6 (3 amplify, 2 dampen), financial development in 5 and sector structure in 8 (6 mixed). A second reader agreed with the moderator type in 86% of a 28-entry sample (kappa 0.83) but with the direction in only 71% (kappa 0.64), so the direction counts are indicative only. Amplification is the commonest finding for human capital and infrastructure, which is consistent with, but does not show, conditional returns (H2).

Thresholds. The studies state 21 thresholds in 14 studies, for the human-capital index,

digital maturity, internet penetration, trade openness, private credit and others. They are in different units, rest on one study each and cannot be pooled. Where two studies address the same variable they do not give comparable values.

Evidence gaps. Figure 6 shows that low-income economies are listed in 11 of 37 output studies, 3 of 14 labour-market studies, 2 of 11 distribution studies and 2 of 4 resource-cost studies. High-income economies are listed in 22 of 37 output studies and 11 of 14 labour-market studies. The historical-wave and AI streams look alike on income-group comparison (39% against 38%) but differ on risk of bias (critical in 9% against 31%) and first sample year (median 2000 against 2012); the AI stream lists a low-income economy in 2 of 13 studies against 18 of 54 (Fisher $p = 0.31$).

Output and productivity	11	11	11	22
Structural change	3	3	3	3
Labour market	3	5	6	11
Poverty and distribution	2	1	3	7
Living standards	4	4	4	6
Resource cost	2	1	2	4
	Low	Lower-middle	Upper-middle	High

Figure 6: Evidence-gap map: studies per outcome family by income groups listed in the sample (a study counts in each group it lists; studies that do not state coverage are not shown).

5 Results of the macro-panel analysis

5.1 Mobile adoption and growth by income group (A1, A2)

Table 4 shows that a 10-subscription increase in mobile adoption per 100 people goes with higher annual growth in all four income groups over the same five-year period. The association is largest in lower-middle income economies (0.47 percentage points, 95% CI 0.16 to 0.77) and about 0.16 in upper-middle and high-income economies; the low-income estimate (0.36) is imprecise (CI -0.05 to 0.76 , 20 economies). With country fixed effects, the lower-middle estimate remains (0.36, $p = 0.03$), and the others move toward 0.13 to 0.20. Extending the sample to 2023 gives the same pattern (lower-middle 0.34, $p < 0.01$; upper-middle 0.05, not significant; Appendix Table 21).

The predetermined specification, which uses the previous period's change, does not reproduce this pattern: the association is positive only in lower-middle income economies (0.26, $p = 0.02$), near zero in upper-middle, and negative in low-income (-0.38 , $p = 0.13$) and high-income (-0.16 , $p = 0.01$) economies, and negative in every income group for internet use (Appendix Table 19). The contemporaneous associations therefore partly reflect growth driving adoption or shared shocks, and they should not be read as a payoff to adoption. Results for fixed broadband rest on extrapolation from very few observations and are not interpreted (Appendix Table 20).

Table 4: Mobile adoption and annual GDP-per-capita growth by income group: change in annual growth (percentage points) per +10 subscriptions per 100 people, five-year periods 1995–2019 (95% CI; standard errors clustered by country).

Income group	Economies	Same period (A1)	Previous period (A2)
Low	20	0.36 [−0.05, 0.76]	−0.38 [−0.86, 0.11]
Lower-middle	42	0.47 [0.16, 0.77]	0.26 [0.03, 0.49]
Upper-middle	51	0.16 [0.04, 0.28]	0.01 [−0.15, 0.16]
High	64	0.16 [0.05, 0.26]	−0.16 [−0.28, −0.03]

5.2 Dose-response and differences inside low-income economies (A3, A6)

Across baseline adoption levels, the association is largest where adoption was below 10 per 100 (0.41) and smaller at 40 to 80 (0.18); it is not monotone, since it is 0.27 above 80 per 100. For low and lower-middle income economies the point estimate above 80 per 100 is the largest (1.37) but imprecise (95% CI −0.01 to 2.74, 35 observations; Appendix Table 23). Inside low and lower-middle income economies, none of initial secondary enrolment, electricity access, private credit or initial income significantly changed the association (all $p > 0.19$; Appendix Table 24); the intervals are wide, and a standard-deviation change in a moderator could change the association by about 0.3 percentage points in either direction.

5.3 Event study around mobile takeoff (A4b)

The first event-study design, stacked cohorts with controls that adopt more than ten years later or never, was not interpretable (Section 3, Appendix H): nearly every economy crosses the threshold in the window, only 7 clean controls existed for the low and lower-middle income group, and that group showed a pre-trend of 12 log points. The replacement design compares each cohort with economies not yet treated at each horizon and adjusts for initial income and earlier growth. It passes the pre-trend rule in all four samples when the takeoff threshold is 10 subscriptions per 100 (mean placebo effect between −0.9 and 0.6 log points, all intervals including zero; Table 5).

Table 5: Event study around the year mobile subscriptions reach 10 per 100 people: effect on log GDP per capita ($\times 100$) relative to the year before takeoff and to economies not yet treated (95% bootstrap interval, 400 draws). Pre-trend: mean of the placebo effects at −5 to −2 years.

Sample	Treated	Pre-trend	Effect at +5 years	Effect at +10 years
All economies	178	−0.0 [−0.6, 0.4]	13.5 [0.6, 36.5]	47.8 [−12.5, 149.1]
Low and lower-middle	66	0.6 [−0.6, 1.6]	−5.9 [−18.6, 31.9]	not estimable
Upper-middle	56	−0.9 [−2.0, 0.3]	11.9 [−8.0, 46.7]	12.2 [−70.1, 118.6]
High	56	−0.4 [−1.2, 0.5]	19.1 [6.2, 36.7]	53.8 [4.2, 152.6]

Economies that reach 10 subscriptions per 100 earlier have higher GDP per capita five years later than comparable economies that reach it later, by about 14 log points (95% CI 0.6 to 36.5). The difference is distinguishable from zero only for high-income economies (19 log points, CI 6.2 to 36.7); the interval for low and lower-middle income economies, −18.6 to +31.9, is wide enough to contain effects of either sign and of the high-income size. Magnitudes at ten years (about 48 log points overall) are implausibly large for an effect of mobile telephony alone. They mainly show that economies reaching the threshold early are on different growth paths from

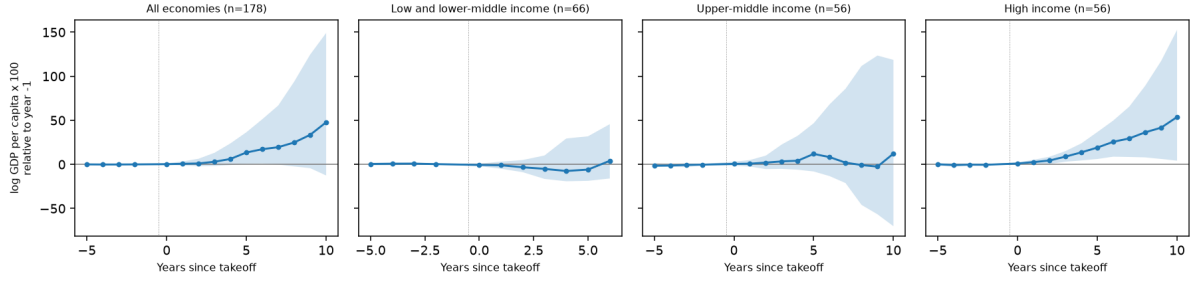


Figure 7: Event-study estimates by income group (takeoff at 10 mobile subscriptions per 100; shaded: 95% bootstrap interval). The low and lower-middle panel stops at six years because fewer than eight not-yet-treated economies exist at longer horizons.

those reaching it later, in ways initial income and earlier growth do not capture, and that at long horizons only the earliest cohorts contribute. The results are similar with a threshold of 25 per 100 (high income: 15.7 at five years, CI 6.5 to 28.8; low and lower-middle: -7.5 , CI -17.6 to 13.3). With a threshold of 50, the pooled sample and the upper-middle sample fail the pre-trend rule and are not interpreted; the low and lower-middle estimate is 4.3 (CI -10.7 to 13.9) and the high-income estimate 8.9 (CI -0.8 to 20.8) (Appendix Tables 25 and 26).

Outcomes beyond income. For samples that pass the pre-trend rule, life expectancy is about 0.9 years higher five years after takeoff in the pooled sample (CI 0.1 to 1.7) and 0.8 years in high-income economies (CI 0.0 to 1.5), with a larger but imprecise estimate for low and lower-middle income economies (2.6 years, CI -0.5 to 6.4). Under-5 mortality is 15 log points lower in low and lower-middle income economies (CI -26.0 to 1.0). Unemployment and the agriculture share fail the pre-trend rule in most samples, and electricity use is not distinguishable from zero (Appendix Table 27). These outcomes have strong secular trends, so we read these as descriptive.

5.4 Local projections (A7)

Local projections give a dynamic profile without a binary treatment (Table 6, Figure 8). For lower-middle income economies, a 10-subscription increase over three years is followed by cumulative growth that is higher by 0.41 percentage points after one year and by 2.0 points after six years (CI 0.8 to 3.2), then 1.7 after eight. For upper-middle and high-income economies, the profile is flat, and the high-income point estimate turns slightly negative (cumulative -0.5 after eight years, CI -1.2 to 0.2). For low-income economies the profile starts positive and turns negative (-4.3 after eight years, CI -9.2 to 0.6), but the intervals include zero except as noted. The placebo (future adoption against past growth) is close to zero in every group, though with a wide interval for low-income economies (-0.75 , CI -2.5 to 1.0).

5.5 Consistency across designs

Table 7 summarises which income group shows a positive association under each design. No group is positive under all four. Lower-middle income economies are positive under three (A1, A2, local projections); under the event study they are pooled with low-income economies and the pooled estimate is null with a negative point estimate. High-income economies are positive under the same-period regression and the event study, but flat or negative under the predetermined

Table 6: Local projections: cumulative growth of GDP per capita (percentage points) h years after a +10 change in mobile subscriptions per 100 over the previous three years (95% CI; country and year fixed effects).

Group	Economies	$h = 1$	$h = 2$	$h = 4$	$h = 8$
Low	24	0.25 [−0.08, 0.58]	0.34 [−0.35, 1.04]	−0.44 [−2.23, 1.35]	−4.31 [−9.19, 0.57]
Lower-middle	47	0.41 [0.11, 0.71]	0.76 [0.25, 1.27]	1.51 [0.69, 2.33]	1.69 [0.32, 3.07]
Upper-middle	58	−0.00 [−0.21, 0.21]	0.10 [−0.19, 0.38]	0.22 [−0.45, 0.88]	0.17 [−0.99, 1.34]
High	74	0.08 [−0.04, 0.19]	0.06 [−0.16, 0.29]	−0.06 [−0.48, 0.35]	−0.51 [−1.23, 0.20]

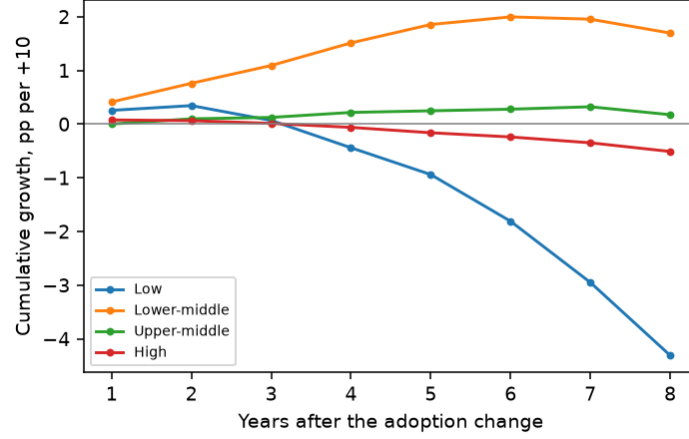


Figure 8: Local-projection profiles by income group.

regression and the local projections. Upper-middle economies are positive only in the same-period regression. Low-income economies are never significantly positive. We conclude that the panel analysis does not support a statement that richer or poorer economies gain more; the answer depends on the design, which is the same message as in the reviewed studies.

Table 7: Direction and significance of the association between mobile adoption and GDP per capita growth, by design and income group (+ positive and significant at 5%, − negative and significant, 0 not significant; sign of the point estimate in parentheses).

Design	Low	Lower-middle	Upper-middle	High
A1 same period	0 (+)	+	+	+
A2 previous period	0 (−)	+	0 (+)	−
Event study (+5 years)	0 (−), low and lower-middle pooled		0 (+)	+
Local projections ($h = 6$)	0 (−)	+	0 (+)	0 (−)

6 Discussion

6.1 What the evidence supports

Four points stand out across the review and the panel analysis. First, the central estimate of the association between ICT-type adoption and output is positive but modest and very heterogeneous across studies (H1), and it shrinks when we allow for small-study bias. GMM estimates are lower than other estimates, but restricting to studies with an identification strategy does not lower the pooled value, so H4 has only mixed support. A partial correlation of 0.19

means that adoption explains a few percent of the residual variation in output, and the bias-corrected estimates are smaller still.

Second, returns look conditional on human capital, financial depth, governance and infrastructure (H2). Amplification is the commonest finding when authors test these conditions. This agrees with the long-run diffusion evidence that the intensity of use, rather than arrival, drives divergence [3], although the studies we synthesise cannot show that changing a moderator would change the return. The threshold results, such as a human-capital index below which the effect of the digital economy is negative in Africa, suggest that the conditions are not marginal but may bind. That reading needs replication, since each threshold comes from one study.

Third, the question of whether poorer or richer economies gain more cannot be answered from the reviewed evidence (H3). The comparisons that exist are few, mostly informal and often from weak designs, and low-income economies are rarely sampled. The measured association in the studies leans toward richer samples, but design differences could produce that lean.

Fourth, the timing of effects matters, but few studies examine it (H5). A study with a five-year lag found a positive effect where the contemporaneous effect was null. In our own analysis, predetermined and contemporaneous specifications give different pictures; the local projections show gains in lower-middle income economies building over about six years, and the event study shows gains in high-income economies emerging after the third year. Any claim about gains from a technology should state the horizon.

6.2 Reconciling the review with the panel analysis

The panel analysis adds a cautious observation. Under a single same-period design, the association between mobile adoption and growth is not smaller in poorer economies: it is largest in lower-middle income economies. This runs against the lean in the reviewed studies toward larger measured gains in richer samples. But the picture depends on the design: the event study finds a distinguishable gain only for high-income economies, and local projections only for lower-middle income economies, so the panel analysis does not settle which group gains more either. The two sources differ in technology (the panel covers mobile subscriptions only), in outcome (growth of GDP per capita) and in design (we use one specification for all groups, while the studies use different specifications by group), so they are not in direct conflict. The predetermined version does not reproduce the association, so we do not take it as evidence that poorer economies gain more. The honest summary is that the two sources do not agree on a difference by income, and neither identifies a causal effect.

6.3 AI

The evidence on AI is too thin and short to be put next to the evidence on earlier waves (H6). The few studies that exist show patterns similar to those for ICT where they test them: larger measured effects in richer economies and in high-skill occupations. But the exposure measures are indices of readiness or occupational exposure, the panels are short, 31% of the AI studies are rated critical, and only two list a low-income economy. A comparison with earlier technologies would require evidence on realised use, which none of the included studies has.

6.4 Implications

For policy, the safest reading is that digital investment is unlikely to pay off equally everywhere and that complementary capabilities matter. The review does not support statements of the form that a given income group should expect a given payoff. Three practical points follow. Evaluations of digital programmes should report effects at several horizons, because contemporaneous estimates can mislead. Programmes in low-income settings should be paired with, or at least monitored against, the complements the literature highlights, such as skills, electricity and finance. And claims about AI should be treated as provisional until studies with measures of realised use and credible designs appear.

6.5 Research agenda

For research, the main need is identification for developing and low-income economies, with instruments or timing designs of the kind used for submarine cables [6], and for AI, measures of realised use rather than readiness. Second, studies should report group-specific effects with a test of the difference, which most studies do not. Third, a registered, two-reviewer systematic review with Scopus and Web of Science searches would reduce the selection concerns listed below. Fourth, the thresholds reported in single studies should be tested for replication across samples.

7 Limitations

- **Selection.** Only studies with a free or supplied full text were read: about 140 of the 247 records sent to full-text assessment are unread, and paywalled studies are over-represented among them. Records from four venues the author judged unreliable were skipped. Search covered OpenAlex, Crossref and NBER only; Scopus, Web of Science and EconLit were not run.
- **Single reviewer.** Screening used one assisted reviewer, so no agreement statistic exists. Extraction was by machine with a second machine pass and then checked by the author by hand, but no second human reviewer verified it independently.
- **Registration.** The protocol was not registered before the search. The certainty of the evidence was not graded (GRADE was not applied).
- **Design quality.** No study has low risk of bias; most treat adoption as exogenous. We report associations, not causal effects. First-pass risk-of-bias ratings are used; nine ratings would change under the second pass.
- **Counting.** Direction counts are not weighted by size or quality and are not effect sizes. Study-level income comparisons mix World Bank income groups, regions and developed-developing splits.
- **Meta-regression.** Eighteen papers; the clustered standard errors are optimistic. Partial correlations rely on an assumed degrees of freedom. Composite indices, levels and growth outcomes, and conditional main effects are mixed in one analysis despite exclusions. Two

Asian studies (302 and 323) share authors, panel and index and are not independent. The sample is not a census, and the low- and middle-income group is small.

- **Language and period.** English only, 1995 onward.
- **Macro-panel analysis.** The World Bank panel analysis gives conditional associations only. Income groups use the current classification; the planned check with initial income was not run; CO2 emissions could not be retrieved. Roughly forty estimates were reported without correction for multiple testing.

8 Conclusion

Across 67 studies, ICT-type adoption is usually associated with higher output, but the association is modest, strongly heterogeneous, shrinks under a correction for small-study bias, is lower in GMM estimates, and is conditional on human capital, financial depth, governance and infrastructure. Whether rich or poor economies gain more is not established, and the evidence on AI is too thin to compare with earlier waves. A World Bank panel analysis finds that mobile adoption goes with faster growth in every income group in a same-period regression, most strongly in lower-middle income economies, but the pattern changes with the design (a staggered event study finds a gain only for high-income economies, and local projections only for lower-middle income economies), so without a causal design it is not a payoff. Credible identification for low- and middle-income economies and realised-use measures for AI are the main gaps.

Declarations

Data and code availability. The protocol, search and screening logs, extraction database, second-pass verification files, meta-regression inputs, World Bank panel and all code are in the project repository: <https://github.com/vinhqdang/Journal-of-Economy-and-Technology>. Full-text PDFs are not shared for copyright reasons.

Ethics. Not applicable; the review uses published studies and public statistics only.

Author contributions (CRediT). [author to complete]

Conflict of interest. [author to complete]

Funding. No external funding was declared in the protocol. [author to confirm]

Use of AI tools. Generative AI tools were used to support this work: to polish the writing, to assist with screening records and extracting data from the included studies (described in the Methods), and to help write and run analysis code. The ideas, the research question and direction, and the choice of methods and algorithms are the author's own. The author reviewed the AI-assisted output and takes full responsibility for the content of this paper.

Registration. [OSF registration pending]

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A Included studies

Table 8: Studies included in the synthesis ($n = 67$). Authors are not listed because the extraction database does not store them; the DOI identifies each record. n.r. = not reported. RoB = overall risk of bias in the first-pass rating (moderate, serious, critical); no study was rated low.

ID	Year	Title	Technology	Econ.	RoB	DOI
10	2020	Financial inclusion, information and communication technology diffusion, and economic growth: a panel data analysis	ICT, Mobile, Internet	41	serious	https://doi.org/10.1080/02681102.2020.1734770
29	2019	The Internet of Things and economic growth in a panel of countries	ICT, Mobile, Internet	82	serious	https://doi.org/10.1080/10438599.2019.1695941
45	2024	Financial inclusion through digitalization and economic growth in Asia-Pacific countries	ICT, Mobile, Internet	30	serious	https://doi.org/10.1016/j.irfa.2024.103596
57	2020	Nexus between energy consumption, information and communications technology, and economic growth: An enquiry into emerging Asian countries	ICT, Mobile, Internet	10	serious	https://doi.org/10.1002/pa.2172
68	2022	Enhancing human development in developing regions: Do ICT and transport infrastructure matter?	ICT, Mobile, Internet	79	serious	https://doi.org/10.1016/j.techfore.2022.121725
71	2022	The ICT, financial development, energy consumption and economic growth nexus in MENA countries: dynamic panel CS-ARDL evidence	ICT, Mobile, Internet	11	serious	https://doi.org/10.1080/00036846.2022.2096861
88	2005	Does the digital divide matter? The role of information and communication technology in cross-country level and growth estimates	ICT, Mobile, Internet	94	serious	https://doi.org/10.1080/1043859042000304043
108	2024	New technologies and jobs in Europe	AI	16	moderate	https://doi.org/10.1093/epolic/eiae058
126	2023	Job creation and destruction in the digital age: Assessing heterogeneous effects across European Union countries	ICT	27	moderate	https://doi.org/10.1016/j.econmod.2023.106405
138	2023	Empirical Analysis of Inclusive Growth, Information and Communication Technology Adoption, and Institutional Quality	ICT, Mobile, Internet	193	serious	https://doi.org/10.3390/economies11040124
140	2015	What are the drivers of total factor productivity in the European Union?	ICT	17	serious	https://doi.org/10.1080/10438599.2015.1067007
162	2024	Impact of digitization on carbon productivity: an empirical analysis of 136 countries	Mobile, Internet	136	serious	https://doi.org/10.1038/s41598-024-55848-2
174	2025	Artificial intelligence and global carbon inequality: Addressing the challenges and opportunities for SDG 10, SDG 12, and SDG 13	AI	67	serious	https://doi.org/10.1016/j.gsf.2025.102072

ID	Year	Title	Technology	Econ.	RoB	DOI
217	2021	Does digital financial inclusion moderate or exacerbate output volatility?	Mobile	40	serious	https://doi.org/10.1080/13504851.2021.1963400
239	2018	What drives labor market polarization in advanced countries? The role of China and technology	ICT	19	serious	https://doi.org/10.1093/icc/dty063
245	2023	The impact of the digital economy on green total factor productivity in Belt and Road countries: the mediating role of energy transition	ICT, Internet	40	serious	https://doi.org/10.3389/fenvs.2023.1213961
268	2023	Informality and aggregate labor productivity growth: Does ICT moderate the relationship?	ICT, Internet	118	serious	https://doi.org/10.1016/j.telpol.2023.102681
269	2024	Economic growth and the proliferation of ICT infrastructures: which causes the other?	ICT, Mobile, Internet	41	serious	https://doi.org/10.1080/1331677x.2024.2310100
276	2025	Does artificial intelligence help in improving human capital based educational development? Evidence from 29 countries	AI	29	serious	https://doi.org/10.1016/j.techsoc.2025.103004
278	2024	Towards ICT diffusion and trade liberalisation on inclusive growth in Sub-Saharan Africa	ICT, Mobile, Internet	48	serious	https://doi.org/10.1007/s10668-023-04355-x
285	2022	ICT and Economic Growth in EU: A macro Level Comparison of Estimated ICT Output Elasticities	ICT	27	moderate	https://doi.org/10.1080/1097198x.2022.2094182
302	2013	Does ICT Participate in Economic Convergence among Asian Countries: Evidence from Dynamic Panel Data Model	ICT, Mobile, Internet	24	serious	https://doi.org/10.12948/issn14531305/17.2.2013.01
320	2022	The Effect of Technology Adoption on Financial Inclusion: A Cross-country Panel Analysis between China and Nigeria	Mobile, Internet	20	serious	https://doi.org/10.24018/ejbmr.2022.7.2.1314
321	2024	Information and communication technology and labour productivity growth: a production-frontier approach	ICT	24	serious	https://doi.org/10.1007/s10479-024-05818-8
323	2014	Total Factor Productivity, Demographic Traits and ICT: Empirical Analysis for Asia	ICT, Mobile, Internet	24	serious	https://doi.org/10.12948/issn14531305/18.1.2014.01
341	2024	The impact of information and communication technologies on export diversification: Evidence from developing countries	ICT, Mobile, Internet	110	moderate	https://doi.org/10.1080/09638199.2024.2419406
358	2023	Digital Divide, Credit Financing and Financial Inclusion: Changing Patterns of Poverty, Employment and Income Inequalities	Mobile, Internet	148	serious	n.a.
392	2018	TECHNOLOGICAL SOURCES OF ECONOMIC GROWTH IN EUROPE AND THE U.S.	ICT	10	serious	https://doi.org/10.3846/20294913.2017.1280557

ID	Year	Title	Technology	Econ.	RoB	DOI
393	2013	The Determinants of Total Factor Productivity in the EU: Insights from Sectoral Data and Common Dynamic Processes	ICT	17	serious	n.a.
407	2015	The Role of ICT in the Productivity of Central and Eastern European Countries: Cross-country Comparison	ICT, Internet	21	serious	n.a.
408	2025	Effects of the digital economy on economic growth in Africa: the role of human capital	ICT, Mobile, Internet	41	serious	https://doi.org/10.1080/23322039.2025.2525485
422	2024	Icts And Labour Productivity Nexus In Developing Countries: Evidence From Panel Estimation Approach	ICT, Mobile, Internet	84	serious	https://doi.org/10.33736/ijbs.6868.2024
441	2024	Artificial Intelligence Readiness and Employment: A Global Panel Analysis	AI	78	critical	https://doi.org/10.24818/18423264/58.4.24.04
442	2025	From Digital Divide to Employment Equity: How Digitalization Affects Women's Involvement in the Workforce in OECD Countries	Mobile, Internet	22	critical	https://doi.org/10.17583/generos.15672
535	2026	The Economic Value of Agentic AI: A Comparative Analysis of Its Impact on Growth and Business Productivity in Developed and Emerging Economies	AI	10	critical	https://doi.org/10.9734/ajrcos/2026/v19i5859
617	2003	Bridging the digital divide: the growth implications of e-commerce for small and developing states	Internet	205	critical	n.a.
707	2026	Digitalisation and structural change: Evidence from cross-country analysis	Mobile, Internet	51	serious	https://doi.org/10.1016/j.jdec.2026.03.003
750	2025	Artificial Intelligence and Labour Markets in Southeast Asia: An Empirical Examination	AI	1	serious	https://doi.org/10.46557/001c.132415
802	2015	Differences in Job De-Routinization in OECD Countries: Evidence from PIAAC	ICT	22	serious	n.a.
803	2020	The Race between Technological Progress and Female Advancement: Changes in Gender and Skill Premia in OECD Countries	ICT	11	moderate	n.a.
879	2026	AI capability and labor productivity in low- and middle-income countries: An absorptive capacity approach	AI, ICT, Mobile, Internet	63	serious	https://doi.org/10.1016/j.technovation.2026.103645
913	2026	Estimation of Firm Labour Productivity and Sales Growth from Artificial Intelligence in Sub-Saharan African Countries	AI	10	critical	https://doi.org/10.12688/f1000research.179912.1

ID	Year	Title	Technology	Econ.	RoB	DOI
952	2026	Twin Transition and Labor Informality in Asia-Pacific: Threshold Effects and Income-Group Heterogeneity	ICT, Internet	17	serious	https://doi.org/10.3390/economies14090398
957	2024	IMPACT OF ICT ON ECONOMIC GROWTH: CASE OF OECD AND MENA COUNTRIES	ICT, Mobile, Internet	32	serious	https://doi.org/10.59671/dga69
988	2023	IMPACT OF DIGITALISATION ON PRODUCTIVITY GROWTH IN EU MEMBER STATES	ICT	27	serious	https://doi.org/10.30525/2661-5169/2023-3-2
1013	2025	The Effect of Artificial Intelligence Investments on Economic Development: Panel Data Analysis	AI	8	serious	https://doi.org/10.17694/bajece.1802500
1025	2005	Testing the impact of ICT in developed countries during 1980-1995: distributional analysis in Solow's growth accounting framework	ICT	15	serious	n.a.
1026	2026	The nonlinear link between AI and energy poverty: Empirical evidence from global data	AI	64	serious	https://doi.org/10.1016/j.jclepro.2026.148590
1031	2025	Exploring the Moderating impact of ICT Infrastructure on Trade Openness-Sectoral Growth Nexus in Sub-Saharan Africa (SSA)	ICT, Mobile, Internet	26	serious	https://doi.org/10.58709/niujhu.v10i3.2267
1192	2026	The Digital Maturity Paradox: The Divergent Impact of Network Readiness and FDI Across Development Tiers	ICT	121	critical	https://doi.org/10.3390/economies14080320
1372	2024	The relationship between inequality and poverty in developing countries: mitigating role of virtual social network and internet access in schools	Internet	57	serious	https://doi.org/10.1108/ijse-09-2023-0695
1419	2017	Mobile banking usage, quality of growth, inequality and poverty in developing countries	Mobile	93	serious	https://doi.org/10.1177/0266666917744006
1428	2021	IMPACT OF MOBILE PHONES AND INTERNET USE ON FINANCIAL INCLUSION: EMPIRICAL EVIDENCE FROM THE EU POST-COMMUNIST COUNTRIES	Mobile, Internet	11	serious	https://doi.org/10.3846/tede.2021.14508
1438	2021	Impact of Economic Integration and Information and Communication Technology on Economic Growth for European Union: Dynamic Panel GMM Approach	ICT, Mobile, Internet	27	serious	https://doi.org/10.5296/ber.v11i4.19064
1448	2023	The effect of information and communication technology on economic growth high-income countries	ICT, Mobile, Internet	10	serious	https://doi.org/10.55493/5002.v13i9.4824

ID	Year	Title	Technology	Econ.	RoB	DOI
1469	2026	Reducing Gender Inequality in Education in Developing Countries: Do Information and Communication Technologies Matter?	ICT, Mobile, Internet	64	serious	https://doi.org/10.1002/sd.71201
1494	2026	Digital transformation and income disparities: A threshold analysis in developing countries	ICT, Mobile, Internet	45	serious	https://doi.org/10.1556/032.2025.00183
1526	2022	Artificial Intelligence and Employment: New Cross-Country Evidence	AI	23	moderate	https://doi.org/10.3389/frai.2022.832736
1564	2025	The Impact of Artificial Intelligence on Employment: A Panel Data Analysis for Selected Countries	AI	29	serious	https://doi.org/10.30784/epfad.1621455
1596	2026	Impact of Generative AI Adoption on Micro and Small Enterprises in Emerging Economies	Other	3	critical	https://doi.org/10.64060/ijpssdv2i12
1645	2025	Cryptocurrency Adoption Patterns and Financial Inclusion Outcomes in Emerging Market Economies	Mobile, Internet, Crypto	15	critical	https://doi.org/10.71086/iajir/v12i4/iajir1244
1652	2025	Crypto-driven growth: A comparative study of Bitcoin and Ethereum on economic growth for multi-country analysis	Crypto	14	critical	https://doi.org/10.32609/j.ruje.11.164511
1670	2019	FINANCIAL INCLUSION, MOBILE PHONE DIFFUSION, AND ECONOMIC GROWTH; EVIDENCE FROM AFRICA	Mobile	32	serious	https://doi.org/10.32479/ijefi.8256
1679	2022	The Employment Effects of Mobile Internet in Developing Countries	Mobile, Internet	14	moderate	n.a.
1713	2012	ICT capital and labour productivity growth: A non-parametric analysis of 14 OECD countries	ICT	14	serious	https://doi.org/10.1016/j.telpol.2011.12.012
1759	2022	ICT capital–skill complementarity and wage inequality: Evidence from OECD countries	ICT	14	moderate	https://doi.org/10.1016/j.labeco.2022.102151
1798	2022	The role of governance in the effect of the internet on financial inclusion in sub-Saharan Africa	Internet	42	serious	https://doi.org/10.2139/ssrn.4315622

B Results by outcome family

Table 9: Studies reporting output and productivity outcomes ($n = 37$). “Directions” counts the study’s results in this family: + positive, – negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
10	ICT, Mobile, Internet	41	2004–2015	GMM	20, 5m	serious
29	ICT, Mobile, Internet	82	2010–2017	FE/RE/OLS	7+, 10	serious
45	ICT, Mobile, Internet	30	2014–2021	FE/RE/OLS	5+	serious

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
57	ICT, Mobile, Internet	10	2000–2017	ARDL/coint./heterog. panel	2+, 1–, 20, 1m	serious
71	ICT, Mobile, Internet	11	1980–2018	ARDL/coint./heterog. panel	3+	serious
88	ICT, Mobile, Internet	94	1983–1997	FE/RE/OLS	4+, 1m	serious
140	ICT	17	1995–2007	ARDL/coint./heterog. panel	1+	serious
217	Mobile	40	2009–2017	GMM	5+, 1m	serious
245	ICT, Internet	40	2006–2021	ARDL/coint./heterog. panel	3–, 2m	serious
268	ICT, Internet	118	1997–2017	FE/RE/OLS	3+, 3–, 30, 2m	serious
269	ICT, Mobile, Internet	41	2000–2019	ARDL/coint./heterog. panel	2+, 10	serious
285	ICT	27	1996–2016	FE/RE/OLS	6+	moderate
302	ICT, Mobile, Internet	24	2000–2010	GMM	2+, 1–	serious
321	ICT	24	1995–2019	Accounting/DEA/other	3+, 1–, 10, 2m	serious
323	ICT, Mobile, Internet	24	2000–2010	GMM	2+	serious
358	Mobile, Internet	148	2008–2021	FE/RE/OLS	1+	serious
392	ICT	10	1980–2010	Accounting/DEA/other	3+, 10	serious
393	ICT	17	1995–2007	ARDL/coint./heterog. panel	3+, 10	serious
407	ICT, Internet	21	1993–2011	FE/RE/OLS	4+, 10, 1m	serious
408	ICT, Mobile, Internet	41	2000–2021	GMM	5+, 1–	serious
422	ICT, Mobile, Internet	84	2000–2019	GMM	5+, 1m	serious
535	AI	10	2015–2024	FE/RE/OLS	7+	critical
617	Internet	205	2000	Accounting/DEA/other	2+	critical
879	AI, ICT, Mobile, Internet	63	2005–2019	GMM	7+, 1m	serious
913	AI	10	2007–2024	FE/RE/OLS	4+, 20, 2m	critical
952	ICT, Internet	17	2005–2022	GMM	2+	serious
957	ICT, Mobile, Internet	32	2010–2023	GMM	3+, 40, 1m	serious
988	ICT	27	2017–2022	FE/RE/OLS	2+, 10	serious
1025	ICT	15	1980–1995	Accounting/DEA/other	20, 2m	serious
1192	ICT	121	2023	FE/RE/OLS	1+, 1–, 10, 1m	critical
1419	Mobile	93	2005–2011	FE/RE/OLS	2m	serious
1438	ICT, Mobile, Internet	27	2002–2019	GMM	2+, 2m	serious
1448	ICT, Mobile, Internet	10	2001–2020	FE/RE/OLS	40	serious
1596	Other	3	2025	FE/RE/OLS	2+	critical
1652	Crypto	14	2015–2023	ARDL/coint./heterog. panel	3+, 2m	critical
1670	Mobile	32	2004–2016	Accounting/DEA/other	1+, 1–, 20, 1m	serious
1713	ICT	14	1995–2005	Accounting/DEA/other	1+, 2–, 10, 1m	serious

Table 10: Studies reporting structural change outcomes ($n = 5$). “Directions” counts the study’s results in this family: + positive, – negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
239	ICT	19	1996–2010	FE/RE/OLS	20	serious
341	ICT, Mobile, Internet	110	1995–2019	GMM	6–	moderate
707	Mobile, Internet	51	1990–2018	GMM	4+, 3–	serious
1031	ICT, Mobile, Internet	26	1991–2023	ARDL/coint./heterog. panel	3+, 5–, 20	serious
1679	Mobile, Internet	14	2000–2017	FE/RE/OLS	2+, 10	moderate

Table 11: Studies reporting labour market outcomes ($n = 14$). “Directions” counts the study’s results in this family: + positive, – negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
108	AI	16	2011–2019	FE/RE/OLS	2+, 1–, 30	moderate

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
126	ICT	27	1995–2019	GMM	4+, 1m	moderate
239	ICT	19	1996–2010	FE/RE/OLS	3+, 10, 1m	serious
358	Mobile, Internet	148	2008–2021	FE/RE/OLS	1+, 1–, 10	serious
441	AI	78	2019–2022	FE/RE/OLS	3+, 20, 1m	critical
442	Mobile, Internet	22	2000–2022	ARDL/coint./heterog. panel	2+, 1m	critical
750	AI	1	2014–2020	FE/RE/OLS	2–, 1m	serious
802	ICT	22	2011–2012	FE/RE/OLS	1+, 3–	serious
803	ICT	11	1980–2008	GMM	2+, 2–, 1m	moderate
952	ICT, Internet	17	2005–2022	GMM	2+, 1–, 40, 1m	serious
1526	AI	23	2010–2019	FE/RE/OLS	2+, 2–, 20, 1m	moderate
1564	AI	29	2017–2021	GMM	2+, 1–	serious
1679	Mobile, Internet	14	2000–2017	FE/RE/OLS	6+, 1–, 10, 1m	moderate
1759	ICT	14	1970–2005	GMM	1+	moderate

Table 12: Studies reporting poverty and distribution outcomes (n = 11). “Directions” counts the study’s results in this family: + positive, – negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
162	Mobile, Internet	136	2000–2020	GMM	1–	serious
174	AI	67	1995–2019	GMM	8+, 1m	serious
276	AI	29	2017–2021	FE/RE/OLS	1+, 1m	serious
278	ICT, Mobile, Internet	48	2005–2020	GMM	5+	serious
358	Mobile, Internet	148	2008–2021	FE/RE/OLS	1+, 3–	serious
802	ICT	22	2011–2012	FE/RE/OLS	10	serious
803	ICT	11	1980–2008	GMM	1+, 1–, 10	moderate
1372	Internet	57	2012–2016	FE/RE/OLS	1+, 2–	serious
1419	Mobile	93	2005–2011	FE/RE/OLS	20, 2m	serious
1494	ICT, Mobile, Internet	45	2000–2023	GMM	1+, 1–, 10, 2m	serious
1759	ICT	14	1970–2005	GMM	3+, 1–, 1m	moderate

Table 13: Studies reporting living standards beyond income outcomes (n = 12). “Directions” counts the study’s results in this family: + positive, – negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
10	ICT, Mobile, Internet	41	2004–2015	GMM	1m	serious
68	ICT, Mobile, Internet	79	1990–2018	GMM	6+, 10, 1m	serious
138	ICT, Mobile, Internet	193	2010–2019	GMM	4+, 1–, 4m	serious
276	AI	29	2017–2021	FE/RE/OLS	7+, 1–, 10, 2m	serious
320	Mobile, Internet	20	2005–2020	FE/RE/OLS	3+, 20, 1m	serious
358	Mobile, Internet	148	2008–2021	FE/RE/OLS	1+	serious
1013	AI	8	2012–2023	ARDL/coint./heterog. panel	2+, 10	serious
1026	AI	64	2000–2019	GMM	1–, 2m	serious
1428	Mobile, Internet	11	1996–2017	ARDL/coint./heterog. panel	3+, 20, 1m	serious
1469	ICT, Mobile, Internet	64	2001–2022	GMM	7–, 1m	serious
1645	Mobile, Internet, Crypto	15	2018–2023	Accounting/DEA/other	1+	critical
1798	Internet	42	2004–2020	GMM	1+, 2–	serious

Table 14: Studies reporting resource cost outcomes ($n = 4$). “Directions” counts the study’s results in this family: + positive, – negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
162	Mobile, Internet	136	2000–2020	GMM	4+, 1m	serious
174	AI	67	1995–2019	GMM	1+	serious
245	ICT, Internet	40	2006–2021	ARDL/coint./heterog. panel	1+	serious
1026	AI	64	2000–2019	GMM	1–, 1m	serious

C Search strategy

Search blocks (protocol)

Four concept blocks were combined with AND and searched in title and abstract. The OpenAlex query below is the exact string that was run (restricted to the economics, econometrics and finance field), and the Crossref and NBER searches used 96 combinations of a technology term and a context term.

- **Block A (technology):** information and communication technolog*, ICT, internet, broadband, mobile phone*, mobile telephon*, smartphone*, digital technolog*, digital economy, artificial intelligence, machine learning, deep learning, generative AI, large language model*, general purpose technolog*, cryptocurrenc*, bitcoin, crypto-asset*, blockchain.
- **Block B (economy level):** countr*, cross-country, nation*, developing econom*, emerging econom*, low-income, middle-income, high-income, income group*, developed econom*, advanced econom*.
- **Block C (outcomes):** growth, productivity, GDP, poverty, inequality, employment, wage*, labor or labour share, structural transformation, financial inclusion, welfare, living standard*, human development.
- **Block D (heterogeneity):** heterogene*, absorptive capacity, complementar*, threshold*, human capital, digital divide, convergence, divergence, technology diffusion, technology adoption, leapfrog*.

OpenAlex query as run (2 October 2026)

```
title_and_abstract.search.exact(("information and communication technology" OR "information and communication technologies" OR ICT OR internet OR broadband OR "mobile phone" OR "mobile phones" OR "mobile telephony" OR "mobile telephone" OR smartphone* OR "digital technology" OR "digital technologies" OR "digital economy" OR "artificial intelligence" OR "machine learning" OR "deep learning" OR "generative AI" OR "large language model" OR "large language models" OR "general purpose technology" OR "general-purpose technology" OR "general purpose technologies" OR cryptocurrenc* OR bitcoin OR "crypto-asset" OR "crypto-assets" OR blockchain) AND (countr* OR "cross-country" OR "cross country" OR nation* OR "developing economy" OR "developing economies" OR "emerging economy" OR "emerging economies" OR low-income OR middle-income OR high-income OR "income group" OR "income groups" OR "developed economy" OR "developed economies" OR "advanced economy" OR "advanced economies") AND (growth OR productivity OR GDP OR poverty OR inequality OR employment OR wage* OR "labor share" OR "labour share" OR "structural transformation" OR "financial inclusion" OR welfare OR "living standard" OR "living standards" OR "human development") AND (heterogene* OR "absorptive capacity" OR complementar* OR threshold* OR "human capital" OR "digital divide" OR convergence OR divergence OR "technology diffusion" OR "technology adoption" OR leapfrog*),from_publication_date:1995-01-01,type:article|preprint,language:en,primary_topic.field.id:20
```

Crossref and NBER searches

Technology terms: internet; broadband; mobile phones; information and communication technology; digital technology; artificial intelligence; generative AI; machine learning; cryptocurrency; bitcoin; technology diffusion; general purpose technology. Context terms: cross-country panel

economic growth; developing countries income groups; digital divide; poverty and inequality countries; productivity emerging economies; employment and wages across countries; absorptive capacity human capital; financial inclusion. All pairs (12 by 8 = 96) were run from 1995.

Records by source

Source	Distinct records
OpenAlex	1,840
Crossref	1,285
NBER working papers	579
arXiv	0 (attempted, not completed)
Scopus, Web of Science, EconLit	0 (not run)

D Eligibility, extraction form and risk-of-bias criteria

Eligibility

Criterion	Include	Exclude
Study design	Quantitative empirical studies and meta-analyses with their own estimation strategy	Opinion pieces, purely conceptual papers, case descriptions without estimation, simulation-only studies
Sample, historical waves	At least 10 economies, or an explicit comparison of income groups	Single-country and few-country studies
Sample, AI	At least 3 economies, or an explicit comparison across income levels	Single-country firm or worker studies are listed separately as supplementary evidence
Exposure	ICT in general, mobile, internet or broadband, smartphones and mobile data, crypto-assets, AI	Technology used as a control only
Outcome	At least one of six families: output and productivity; structural change; labour market; poverty and distribution; living standards beyond income; resource cost	Adoption or diffusion as the only outcome (kept as context)
Period and language	1995 to the search date; English	Earlier; other languages
Status	Peer-reviewed articles and working papers from named series; preprints flagged	Conference abstracts, blog posts, reports without methods

Exclusion codes at full text: E1 too few economies and no income-group comparison; E2 exposure not eligible; E3 no outcome in the six families; E4 conceptual, review, or no own estimation; DUP duplicate or earlier version of another paper.

Extraction form (frozen before extraction)

For each study: decision and coded reason; stream (historical waves, AI or both); technology and exposure measure; sample (number of economies, groups, income coverage, years, unit); design (estimator, identification strategy, controls); results by outcome family (indicator, direction, estimate, uncertainty, horizon, locator and quote); heterogeneity findings (moderator, finding, threshold, locator); results by income group; number of specifications; whether the headline estimate is the preferred specification; risk-of-bias ratings with reasons; venue concern; free-text notes. Every extracted result carries a page or table locator and a verbatim quote of at most 25 words.

Risk-of-bias domains and rating guide

1. Confounding and reverse causation: is adoption treated as exogenous, and is there a credible instrument or timing design?
2. Exposure measurement: does the adoption measure capture use, or only access or investment?
3. Selection of countries and years: missing-data patterns and survivorship.
4. Outcome measurement: quality of the national statistics used.
5. Specification and researcher degrees of freedom: number of specifications and sensitivity.
6. Selective reporting.

Rating guide: *low* = credible design for the claim; *moderate* = standard panel design with reasonable controls but exogeneity not established; *serious* = causal language with no strategy for reverse causation, or a measure that does not capture the exposure; *critical* = the conclusions cannot be trusted from this design.

E Meta-regression inputs

Table 15: Estimates used in the meta-regression (103 estimates from 18 studies): partial correlation (PCC) and its standard error, computed from the printed coefficient and uncertainty with degrees of freedom equal to observations minus 10.

Estimate	Role	Technology	Obs.	t	PCC	SE
10-1	headline	mobile	395	5.82	0.284	0.049
10-2	baseline	mobile	395	-7.46	-0.355	0.048
10-3	headline	internet	392	3.31	0.167	0.050
10-4	baseline	internet	392	-4.40	-0.220	0.050
10-5	baseline	telecom	397	-0.15	-0.008	0.051
10-6	baseline	other (ICT goods impor	396	-0.36	-0.018	0.051
29-1	headline	other (cellular IoT co	574	2.56	0.107	0.042
29-10	robustness	other (cellular IoT co	450	6.00	0.275	0.046
29-11	robustness	other (cellular IoT co	252	3.33	0.210	0.063
29-12	robustness	other (cellular IoT co	378	2.86	0.147	0.052
29-13	robustness	other (cellular IoT co	574	2.22	0.093	0.042
29-14	robustness	other (cellular IoT co	574	2.22	0.093	0.042
29-2	baseline	other (cellular IoT co	574	2.60	0.109	0.042

Estimate	Role	Technology	Obs.	t	PCC	SE
29-3	robustness	other (cellular IoT co	410	2.40	0.119	0.050
29-4	robustness	other (cellular IoT co	410	2.21	0.110	0.050
29-5	subsample	other (cellular IoT co	252	1.91	0.122	0.064
29-6	subsample	other (cellular IoT co	252	1.70	0.109	0.064
29-7	subsample	other (cellular IoT co	322	1.92	0.108	0.056
29-8	subsample	other (cellular IoT co	322	1.92	0.108	0.056
29-9	robustness	other (cellular IoT co	630	5.00	0.197	0.039
45-1	headline	other (digital financi	90	5.00	0.488	0.098
45-2	subsample	other (digital financi	90	7.00	0.616	0.088
45-3	subsample	other (digital financi	90	4.00	0.408	0.102
45-4	robustness	other (digital financi	90	5.00	0.488	0.098
45-5	robustness	other (digital financi	90	3.09	0.327	0.106
71-1	headline	ICT	429	1.67	0.081	0.049
71-2	baseline	ICT	429	1.82	0.089	0.049
71-3	robustness	ICT	429	3.55	0.171	0.048
88-10	robustness	computers	181	13.19	0.710	0.054
88-7	robustness	ICT (composite)	465	2.15	0.100	0.047
88-8	robustness	computers	293	14.29	0.647	0.045
88-9	robustness	ICT (composite)	465	1.92	0.090	0.047
140-1	headline	ICT (ICT capital servi	451	3.19	0.150	0.047
140-2	baseline	ICT (ICT capital servi	451	2.40	0.114	0.047
245-1	headline	digital economy	628	-1.90	-0.076	0.040
245-10	subsample	digital economy	373	-3.55	-0.183	0.052
245-11	subsample	digital economy	239	-4.45	-0.282	0.063
245-12	robustness	digital economy	528	-5.74	-0.245	0.043
245-13	robustness	digital economy	605	-2.04	-0.083	0.041
245-14	robustness	digital economy	587	-5.90	-0.239	0.040
245-2	baseline	digital economy	628	-1.97	-0.079	0.040
245-3	robustness	digital economy	587	-4.11	-0.169	0.041
245-4	robustness	digital economy	605	-7.56	-0.296	0.039
245-5	robustness	digital economy	604	-6.12	-0.244	0.040
245-6	subsample	digital economy	306	-3.25	-0.186	0.057
245-7	subsample	digital economy	94	1.56	0.168	0.108
245-8	subsample	digital economy	205	-2.23	-0.158	0.071
245-9	subsample	digital economy	16	-1.66	-0.561	0.338
268-1	headline	other (ICT capital dee	2374	2.36	0.049	0.021
268-10	subsample	other (ICT capital dee	791	2.18	0.078	0.036
268-11	subsample	other (ICT capital dee	791	2.12	0.075	0.036
268-12	subsample	other (ICT capital dee	791	2.60	0.093	0.036
268-2	robustness	other (ICT capital dee	2374	3.23	0.066	0.021
268-3	robustness	other (ICT capital dee	2374	2.17	0.045	0.021
268-4	robustness	other (ICT capital dee	2374	2.90	0.060	0.021
268-5	subsample	other (ICT capital dee	1583	1.73	0.044	0.025
268-6	subsample	other (ICT capital dee	1583	2.46	0.062	0.025
268-7	subsample	other (ICT capital dee	1583	1.62	0.041	0.025
268-8	subsample	other (ICT capital dee	1583	2.00	0.050	0.025
268-9	subsample	other (ICT capital dee	791	2.67	0.095	0.036
285-1	headline	ICT	558	6.57	0.270	0.041
285-10	subsample	ICT	238	6.53	0.397	0.061
285-11	subsample	ICT	320	3.35	0.187	0.056
285-12	subsample	ICT	320	3.24	0.181	0.056
285-2	baseline	ICT	567	5.80	0.239	0.041
285-3	robustness	ICT	558	7.14	0.292	0.041

Estimate	Role	Technology	Obs.	t	PCC	SE
285-4	robustness	ICT	558	5.75	0.239	0.041
285-5	robustness	ICT	558	3.35	0.142	0.042
285-6	robustness	ICT	558	2.64	0.112	0.042
285-7	robustness	ICT	567	6.53	0.267	0.041
285-8	robustness	ICT	567	4.96	0.206	0.041
285-9	subsample	ICT	238	4.74	0.299	0.063
302-1	headline	ICT (composite ICT Mat	216	2.29	0.157	0.069
323-1	headline	ICT (composite index,	216	2.48	0.170	0.069
393-1	headline	ICT	1106	1.97	0.060	0.030
393-2	baseline	ICT	1322	1.37	0.038	0.028
393-3	robustness	ICT	1322	2.97	0.082	0.028
393-4	robustness	ICT	1158	2.37	0.070	0.029
393-5	robustness	ICT	1158	1.50	0.044	0.029
407-1	headline	ICT	399	12.80	0.544	0.043
407-10	robustness	ICT	98	0.09	0.009	0.107
407-11	robustness	ICT	112	-2.00	-0.194	0.097
407-12	robustness	ICT	56	-1.52	-0.218	0.144
407-13	robustness	internet	56	3.00	0.405	0.135
407-2	subsample	ICT	133	7.50	0.560	0.075
407-3	subsample	ICT	266	2.23	0.138	0.062
407-5	subsample	internet	133	5.00	0.411	0.082
407-7	robustness	ICT	49	0.82	0.130	0.159
407-8	robustness	ICT	56	5.09	0.600	0.118
407-9	robustness	ICT	28	0.12	0.028	0.236
408-7	robustness	broadband	779	-2.00	-0.072	0.036
617-1	headline	internet (internet hos	126	6.41	0.511	0.080
617-2	robustness	internet (internet hos	126	6.17	0.497	0.081
952-1	headline	digital economy (compo	198	1.70	0.123	0.072
988-1	headline	digital economy (DESI	168	0.00	0.000	0.080
988-2	robustness	digital economy (DESI	168	0.00	0.000	0.080
988-3	robustness	digital economy (DESI	168	0.00	0.000	0.080
988-4	robustness	digital economy (DESI	168	0.00	0.000	0.080
988-6	robustness	digital economy (DESI	140	0.00	0.000	0.088
988-7	robustness	digital economy (DESI	140	0.00	0.000	0.088
988-8	robustness	digital economy (DESI	168	0.00	0.000	0.080
1438-1	headline	internet	138	2.20	0.191	0.087
1438-2	baseline	mobile	138	1.80	0.157	0.087

F Second-pass check

Table 16: Second-pass check by study: results confirmed (C), corrected (X) or unverifiable (U), heterogeneity entries corrected, and the second reader’s view of the overall risk-of-bias rating.

ID	C	X	U	Het. X	RoB view
10	7	1	0	0	agree
29	8	0	0	0	agree
45	5	0	0	0	agree
57	5	1	0	0	too lenient
68	7	1	0	0	agree
71	not second-passed				

ID	C	X	U	Het. X	RoB view
88	5	0	0	1	agree
108	6	0	0	1	agree
126	5	0	0	0	agree
138	9	0	0	1	agree
140	1	0	0	0	agree
162	6	0	0	0	agree
174	10	0	0	2	agree
217	6	0	0	0	agree
239	7	0	0	0	agree
245	6	0	0	0	agree
268	10	1	0	0	too harsh
269	3	0	0	0	agree
276	13	0	0	0	agree
278	not second-passed				
285	6	0	0	0	agree
302	3	0	0	0	too harsh
320	6	0	0	1	agree
321	6	1	0	0	agree
323	2	0	0	0	too harsh
341	not second-passed				
358	7	2	0	0	agree
392	3	1	0	0	agree
393	4	0	0	0	agree
407	6	0	0	0	agree
408	6	0	0	0	too harsh
422	5	1	0	0	agree
441	6	0	0	0	agree
442	0	2	1	0	agree
535	8	0	0	0	agree
617	2	0	0	1	agree
707	7	0	0	1	too harsh
750	2	0	1	0	agree
802	5	0	0	1	agree
803	7	1	0	0	agree
879	7	1	0	0	agree
913	8	0	0	0	agree
952	10	0	0	0	agree
957	7	0	1	0	too lenient
988	3	0	0	1	agree
1013	3	0	0	0	agree
1025	4	0	0	0	agree
1026	4	1	0	0	agree
1031	9	1	0	1	too lenient
1192	4	0	0	0	agree
1372	3	0	0	0	agree
1419	6	0	0	0	agree
1428	4	2	0	0	agree
1438	4	0	0	0	agree
1448	4	0	0	0	agree
1469	8	0	0	0	too harsh
1494	5	0	0	1	agree
1526	7	0	0	0	agree
1564	3	0	0	0	agree

ID	C	X	U	Het. X	RoB view
1596	2	0	0	0	agree
1645	1	0	0	0	agree
1652	4	1	0	0	agree
1670	3	0	2	0	agree
1679	12	0	0	0	agree
1713	5	0	0	1	agree
1759	6	0	0	0	agree
1798	2	1	0	0	agree

G Macro-panel data and additional results

Variables

Table 17: World Bank WDI indicators used.

Role	Variable	WDI code
Technology	Mobile cellular subscriptions per 100 people	IT.CEL.SETS.P2
Technology	Individuals using the internet, % of population	IT.NET.USER.ZS
Technology	Fixed broadband subscriptions per 100 people	IT.NET.BBND.P2
Outcome	GDP per capita, constant 2015 US\$	NY.GDP.PCAP.KD
Outcome	Life expectancy at birth	SP.DYN.LE00.IN
Outcome	Under-5 mortality rate	SH.DYN.MORT
Outcome	Unemployment, % of labour force	SL.UEM.TOTL.ZS
Outcome	Agriculture, value added % of GDP	NV.AGR.TOTL.ZS
Outcome	Electric power consumption per capita	EG.USE.ELEC.KH.PC
Control or moderator	School enrolment, secondary, gross	SE.SEC.ENRR
Control	Trade, % of GDP	NE.TRD.GNFS.ZS
Moderator	Domestic credit to private sector, % of GDP	FS.AST.PRVT.GD.ZS
Moderator	Access to electricity, % of population	EG.ELC.ACCS.ZS
Other	Population, total	SP.POP.TOTL

Income groups follow the World Bank’s current classification of 217 economies (25 low, 47 lower-middle, 59 upper-middle and 86 high income; the analysis sample is smaller because of missing data). Aggregates were removed. CO₂ emissions per capita could not be retrieved from the API and are not analysed.

Table 18: Internet use and growth, same period (A1): change in annual growth (pp) per +10 points of users.

Income group	Economies	Observations	Effect of +10 on annual growth (pp)
Low	18	53	-0.916 [-3.241, 1.409] (p=0.44)
Lower-middle	42	158	1.185 [0.348, 2.023] (p=0.01)
Upper-middle	51	184	0.370 [-0.040, 0.779] (p=0.08)
High	60	223	-0.089 [-0.340, 0.162] (p=0.49)

Table 19: Internet use and growth, previous period (A2).

Income group	Economies	Observations	Effect of +10 on annual growth (pp)
Low	19	43	-3.861 [-7.810, 0.088] (p=0.06)
Lower-middle	41	127	-0.399 [-1.202, 0.405] (p=0.33)
Upper-middle	49	150	-0.209 [-0.582, 0.165] (p=0.27)
High	60	210	-0.306 [-0.509, -0.102] (p=0.00)

Table 20: Fixed broadband and growth, same period (A1). Not interpreted: extrapolation from few observations.

Income group	Economies	Observations	Effect of +10 on annual growth (pp)
Low	17	34	-21.426 [-27.860, -14.993] (p=0.00)
Lower-middle	40	99	5.846 [2.454, 9.237] (p=0.00)
Upper-middle	47	122	1.830 [0.623, 3.036] (p=0.00)
High	62	192	-0.057 [-0.589, 0.474] (p=0.83)

Table 21: Sensitivity: mobile adoption and growth, same period (A1), sample extended to 2023 (last period 2015–23).

Income group	Economies	Observations	Effect of +10 on annual growth (pp)
Low	20	60	0.329 [-0.123, 0.782] (p=0.15)
Lower-middle	42	166	0.344 [0.168, 0.521] (p=0.00)
Upper-middle	51	186	0.053 [-0.048, 0.154] (p=0.31)
High	61	227	0.134 [0.031, 0.238] (p=0.01)

Table 22: Sensitivity: previous-period specification (A2), sample extended to 2023.

Income group	Economies	Observations	Effect of +10 on annual growth (pp)
Low	19	54	-0.408 [-0.870, 0.054] (p=0.08)
Lower-middle	42	143	0.164 [-0.016, 0.344] (p=0.07)
Upper-middle	51	160	-0.047 [-0.195, 0.101] (p=0.53)
High	62	217	-0.187 [-0.302, -0.072] (p=0.00)

Table 23: Dose-response (A3): change in annual growth (pp) per +10 mobile subscriptions per 100, by baseline adoption level.

Sample	Baseline bin	Obs	Effect of +10 on annual growth (pp)
All economies	<10	186	0.405 [0.249, 0.561] (p=0.00)
All economies	10-40	97	0.281 [0.167, 0.395] (p=0.00)
All economies	40-80	123	0.179 [0.059, 0.298] (p=0.00)
All economies	>80	249	0.271 [0.119, 0.423] (p=0.00)
Low and lower-middle income	<10	109	0.509 [0.171, 0.846] (p=0.00)
Low and lower-middle income	10-40	36	0.359 [0.106, 0.612] (p=0.01)
Low and lower-middle income	40-80	53	0.453 [0.080, 0.826] (p=0.02)
Low and lower-middle income	>80	35	1.367 [-0.005, 2.739] (p=0.05)
Upper-middle and high income	<10	77	0.234 [0.013, 0.455] (p=0.04)
Upper-middle and high income	10-40	61	0.233 [0.093, 0.372] (p=0.00)
Upper-middle and high income	40-80	70	0.120 [-0.008, 0.247] (p=0.07)
Upper-middle and high income	>80	214	0.159 [0.052, 0.266] (p=0.00)

Table 24: Differences inside low and lower-middle income economies (A6).

Moderator (at start of period)	Interaction with +10 mobile (pp growth per SD)	Obs
secondary enrolment	0.136 [-0.205, 0.478] (p=0.43)	233
electricity access	0.024 [-0.301, 0.348] (p=0.89)	224
private credit	-0.074 [-0.295, 0.147] (p=0.51)	204
log GDP per capita	0.237 [-0.128, 0.602] (p=0.20)	233

Table 25: Event study with takeoff at 25 mobile subscriptions per 100 (log GDP per capita $\times 100$; 400 bootstrap draws).

Sample	Treated	Pool	Pre-trend: mean of effects at -5...-2 [95% CI]	Passes pre-trend rule	Effect at
All economies	188	188	-0.2 [-0.7, 0.3]	yes	6.1 [1.8,
Low and lower-middle income	66	188	0.2 [-0.7, 1.4]	yes	-0.5 [-6.1,
Upper-middle income	56	188	-0.9 [-2.3, 0.6]	yes	7.5 [-0.5,
High income	66	188	0.0 [-1.1, 0.6]	yes	9.3 [3.9,

Table 26: Event study with takeoff at 50 mobile subscriptions per 100. Samples that fail the pre-trend rule are not interpreted.

Sample	Treated	Pool	Pre-trend: mean of effects at -5...-2 [95% CI]	Passes pre-trend rule	Effect at
All economies	176	188	-0.9 [-1.4, -0.3]	no	4.6 [-0.1,
Low and lower-middle income	57	188	-0.3 [-1.6, 1.0]	yes	2.1 [-6.6,
Upper-middle income	53	188	-1.9 [-3.5, -0.3]	no	5.3 [-1.6,
High income	66	188	-0.6 [-1.5, 0.2]	yes	4.9 [0.2,

Table 27: Event study, secondary outcomes (takeoff at 10 per 100): effects at five and ten years. Rows marked fails do not pass the pre-trend rule and are not interpreted.

Outcome	Sample	Treated	Pre-trend mean [95% CI]	Effect at +5
Life expectancy (years)	All economies	178	0.05 [-0.00, 0.08] (passes)	0.86 [0.10, 1.69]
Life expectancy (years)	Low and lower-middle income	66	0.03 [-0.07, 0.11] (passes)	2.61 [-0.49, 6.33]
Life expectancy (years)	Upper-middle income	56	0.05 [-0.03, 0.13] (passes)	0.63 [-0.30, 1.77]
Life expectancy (years)	High income	56	0.02 [-0.08, 0.18] (passes)	0.81 [0.02, 1.54]
Under-5 mortality (log x 100)	All economies	178	-0.09 [-0.35, 0.28] (passes)	-4.71 [-13.90, 1.69]
Under-5 mortality (log x 100)	Low and lower-middle income	66	-0.20 [-0.72, 0.54] (passes)	-15.08 [-26.03, -4.13]
Under-5 mortality (log x 100)	Upper-middle income	56	0.02 [-0.57, 0.53] (passes)	-2.97 [-21.71, 15.77]
Under-5 mortality (log x 100)	High income	56	0.21 [-0.47, 0.65] (passes)	-4.01 [-12.47, 4.45]
Unemployment (pp)	All economies	178	0.11 [0.01, 0.21] (fails)	-0.22 [-1.24, 0.80]
Unemployment (pp)	Low and lower-middle income	66	-0.16 [-0.30, -0.03] (fails)	n/a
Unemployment (pp)	Upper-middle income	56	0.54 [0.23, 0.74] (fails)	1.34 [-1.33, 2.99]
Unemployment (pp)	High income	56	0.01 [-0.27, 0.37] (passes)	-1.14 [-2.34, 0.06]
Agriculture share of value added (pp)	All economies	178	0.27 [0.02, 0.62] (fails)	-3.56 [-6.14, -1.00]
Agriculture share of value added (pp)	Low and lower-middle income	66	0.34 [-0.22, 1.05] (passes)	n/a
Agriculture share of value added (pp)	Upper-middle income	56	0.36 [-0.05, 0.74] (passes)	-4.29 [-8.59, 0.01]
Agriculture share of value added (pp)	High income	56	0.16 [-0.19, 0.52] (passes)	-3.24 [-5.71, -0.77]
Electric power use per capita (log x 100)	All economies	178	0.01 [-1.09, 1.04] (passes)	10.26 [-3.84, 24.36]
Electric power use per capita (log x 100)	Low and lower-middle income	66	-0.44 [-2.49, 1.76] (passes)	n/a
Electric power use per capita (log x 100)	Upper-middle income	56	-0.46 [-1.98, 2.11] (passes)	2.12 [-17.66, 21.90]
Electric power use per capita (log x 100)	High income	56	0.43 [-0.98, 1.21] (passes)	14.04 [-1.25, 29.33]

H Protocol amendments

1. (2 Oct 2026) The OpenAlex query was written with explicit phrase variants and restricted to the economics field, because an unrestricted query returned about 7,600 records, too many for one screener.
2. (2 Oct) Four holding categories were added at the abstract stage: context, environmental outcome only, reviews for backward citation search, and supplementary micro evidence.
3. (2 Oct) arXiv was attempted but not completed. Scopus, Web of Science and EconLit were not searched.
4. (2 Oct) Automated stage-1 rules were added before manual reading, with a single assisted screener and a verification sample of 90 decisions.
5. (2 Oct) Included records were ranked in two tiers to decide which paywalled studies to request first.
6. (2 Oct) Records with no free full text and not on the short request list were kept as an abstract-level evidence map, excluded from pooled estimates and from risk-of-bias ratings.
7. (3 Oct) Studies whose only available venue the author judged unreliable were not read and not used.
8. (3 Oct) Three studies (298, 878 and 1495) were moved from include to exclude after a second-pass check. Five regression estimates were excluded from the meta-regression after the same check.
9. (3 Oct) The event-study design in the macro-panel plan was replaced by a staggered design with not-yet-treated controls and covariate adjustment after the first design failed its pre-trend checks; local projections were added.

I PRISMA 2020 checklist

Table 28: PRISMA 2020 checklist: where each item is reported. Section numbers refer to this paper.

Item	Topic	Requirement	Reported	Where
1	Title	Identifies the report as a systematic review	Yes	Title
2	Abstract	Structured abstract	Partly	Abstract (not the full PRISMA abstract checklist)
3	Rationale	Rationale in context of existing knowledge	Yes	Introduction; Conceptual background
4	Objectives	Explicit objectives and questions	Yes	Introduction; Conceptual background
5	Eligibility criteria	Inclusion and exclusion criteria	Yes	Methods (eligibility); Appendix D
6	Information sources	Databases, dates, last search	Partly	Methods (sources); Scopus, Web of Science and EconLit not searched
7	Search strategy	Full strategy for all sources	Yes	Appendix C
8	Selection process	How records were screened, number of reviewers	Partly	Methods (screening); one assisted reviewer
9	Data collection process	Methods of extraction, number of reviewers	Partly	Methods (extraction, second pass); machine extraction with machine second pass
10	Data items	Outcomes and other variables	Yes	Methods (eligibility); Appendix D
11	Risk of bias	Tools and reviewers	Partly	Methods (risk of bias); adapted tool, single rater
12	Effect measures	Measures used for each outcome	Yes	Methods (meta-regression)
13a-f	Synthesis methods	Eligibility for synthesis, methods, heterogeneity, sensitivity	Partly	Methods (synthesis, meta-regression); leave-one-out and quality-restricted analyses in Results (meta-regression)
14	Reporting bias assessment	Methods to assess reporting bias	Yes	Methods and Results (meta-regression: FAT-PET, PEESE)
15	Certainty assessment	Methods to assess certainty (e.g. GRADE)	No	Not done; listed as a limitation
16	Study selection results	Flow diagram, excluded studies	Yes	Results (study selection), Figure 1; reasons coded
17	Study characteristics	Cite and present characteristics	Yes	Results (characteristics), Appendix A
18	Risk of bias in studies	Assessments for each study	Yes	Results (risk of bias), Appendix A
19	Results of individual studies	Summary and effect estimates	Partly	Appendices B and E; no forest plot for all outcomes
20	Results of syntheses	Results, heterogeneity, sensitivity	Yes	Results (all synthesis subsections)
21	Reporting biases	Assessments of reporting bias	Yes	Results (meta-regression)
22	Certainty of evidence	Certainty for each outcome	No	Not done
23	Discussion	Interpretation, limitations, implications	Yes	Discussion; Limitations
24	Registration and protocol	Registration number, protocol access	Partly	Protocol in repository; not registered
25	Support	Funding and sponsors	Yes	Declarations (author to confirm)
26	Competing interests	Declared	No	Author to complete

Item	Topic	Requirement	Reported	Where
27	Availability of data and code	Public availability	Yes	Declarations; repository